Arbitrary-Direction SAR Ship Detection Method for Multiscale Imbalance

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***Abstract*— The arbitrary-oriented ship detection in synthetic aperture radar (SAR) imagery remains especially challenging due to multiscale imbalance and the characteristics of SAR imaging, a problem that is more pronounced than in optical ship detection. Unlike optical images, SAR data often lack rich textural and color cues, instead exhibiting nonuniform scattering, speckle noise, and nonstandard elliptical ship shapes, all of which make robust feature extraction and bounding box regression significantly more difficult across different scales. To address these unique SAR-specific challenges, this article proposes the multiscale dynamic feature fusion network (MSDFF-Net) aims to alleviate multiscale imbalance in three main ways. First, a multiscale large-kernel convolutional block (MSLK-Block) integrates large-kernel convolutions with partitioned heterogeneous operations to enhance the multiscale feature representation, tackling wide-ranging ship sizes under noisy conditions. Second, a dynamic feature fusion block (DFF-Block) handles scale-based feature utilization imbalance by adaptively balancing spatial and channel information, thereby reducing interference from clutter and strengthening discrimination for diverse-scale ships. Third, we propose the Gaussian probability distribution (GPD) loss function, which models ships' elliptical scattering properties and mitigates regression loss imbalance for targets of varying scales and orientations. Experimental evaluations on the R-SSDD, R-HRSID, and CEMEE datasets demonstrate that MSDFF-Net reaches top-tier performance standards, outperforming 21 existing deep learning (DL)-based SAR ship detectors. Specifically, MSDFF-Net achieves 93.95% precision, 94.72% recall, 91.55% mean average precision (mAP), 94.33% $F1$-score, and 135.79 frames/s (FPS) on the R-SSDD dataset, with a parameter size of only 8.94 M. In addition, MSDFF-Net exhibits strong transferability across large-scale SAR images, making it suitable for real-world deployment. The code and datasets can be accessed publicly at: https://github.com/SZZ-SXM/MSDFF-Net**

***Index Terms*— Arbitrary-oriented detection, large-kernel network, multiscale imbalance, synthetic aperture radar (SAR).**

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I. Introduction

SYNTHETIC aperture radar (SAR) is a proactive system for wide-area ground observation that utilizes radar platforms, such as missiles, aircraft, and satellites., to transmit and receive radio waves while in motion to acquire high-resolution imagery of terrestrial or maritime targets [1], [2], [3], [4]. Owing to its distinctive imaging mechanism, SAR is unaffected by time and weather conditions, enabling large-scale, real-time, and continuous ground observation. In addition, SAR can also boast strong penetration capabilities and high-resolution imaging ability. Consequently, SAR has significant utilizations in several disciplines, particularly disaster management, resource exploration, urban planning, and ocean monitoring [5], [6], [7], [8], [9], [10], [11].

As the SAR imaging technology continues to evolve, SAR images have been widely used for the detection and recognition of the ship targets, which is of great significance for maritime security and rights protection [12], [13], [14]. Traditional SAR ship detection techniques primarily rely on signal processing and image analysis methods, including but not limited to edge detection, threshold segmentation, and super-pixel detectors [15], [16], [17]. Although the traditional SAR ship target detection technology has promoted the development of marine target detection to a certain extent, it also has obvious limitations and challenges. First, these traditional methods offer poor environmental adaptability and limited capability for suppressing interference, such as sea clutter, shadows, and sidelobe echoes. Second, their parameter selection heavily relies on the operator's experience and may need frequent adjustments across different scenarios. In addition, traditional methods have limited feature representation abilities. Compared with popular deep learning (DL) methods, these conventional approaches exhibit lower automation when processing large-scale datasets and require more manual intervention overall.

Unlike traditional detection approaches, DL-based methods can more effectively cope with complex environments or backgrounds. At the same time, it overcomes the time-consuming and labor-intensive shortcomings of feature design in traditional image interpretation and has more advantages in target detection and recognition, which greatly promotes the enhancement of target detection technology. A variety of DL-based target detection methods have been developed successively [18], [19], [20], [21]. Influenced by the rapid development of general target detection technology, methods

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for detecting ships in SAR images that utilize DL have ushered in another round of development climax. A variety of techniques for SAR ship detection grounded in the optimization of the general target detection framework have been proposed. They have carried out research around the complex scenes of SAR images and the large-scale differences of different targets and have achieved great success. At the same time, in order to more effectively obtain the meaningful features of ship targets in complex scenes, the oriented bounding boxes (OBBs) detection algorithms [22], [23], [24], [25], [26] was derived based on the horizontal bounding boxes (HBBs) detection algorithms [27], [28], [29], [30], [31]. Through the design of the rotation invariant feature extraction module, these methods can better characterize the arbitrary direction characteristics of ships.

Despite recent advancements, DL-based methods for arbitrary-oriented SAR ship detection continue to face key challenges related to multiscale imbalance. First, traditional CNNs lack the global modeling capability essential for representing multiscale ship targets in complex SAR imagery. Unlike optical images, SAR images inherently contain less information, and the extensive use of feature extraction modules can exacerbate object scale imbalance. Furthermore, current SAR ship detection methods rely on generic pyramid structures for feature fusion, overlooking the varying contributions of spatial and channel features under different scales. We identify this issue as a feature utilization imbalance in SAR ship detection. In addition, many oriented detection strategies simply add an angled branch to compute an a intersection over union (IoU) distance loss for angle regression, combining it with classification loss in a multitask framework. However, this fails to address inherent feature learning constraints that do not align well with SAR targets, leading to significant regression loss imbalance [32].

To address multiscale imbalance in feature extraction, feature imbalance in fusion, and regression loss imbalance under SAR-specific conditions, a novel anchor-free arbitrary-oriented SAR ship detection network is proposed, named multiscale dynamic feature fusion network (MSDFF-Net). Building on the validated effectiveness of large-kernel convolutions, we introduce a multiscale large-kernel convolutional block (MSLK-Block) that combines large-kernel operations with a multiscale partitioning mechanism. This design strengthens the network's ability to represent variously sized ship targets amid nonuniform scattering and speckle noise. In the feature fusion stage, a dynamic feature fusion block (DFF-Block) is proposed, inspired by dynamic mechanisms, which enhances the correlation between spatial and channel dimensions during feature fusion, improving its effectiveness. This mechanism enhances the feature correlation across scales and suppresses background interference, improving ship discrimination in complex SAR scenes. To mitigate the scale regression loss imbalance, we rethink the traditional OBB regression paradigm. Rather than directly regressing OBBs, we introduce Gaussian bounding box (GBB) fitting, which better models the elliptical distribution characteristics of ship targets in SAR imagery. As illustrated in Fig. 1, we compare different annotation methods across three SAR ship detection datasets [33], [34], [35] (see Section IV-A for details), including red representing HBBs, blue representing OBBs, and green representing green. During loss regression, the distance B_d defined by the Bhattacharyya measure among the Gaussian probability distributions (GPDs) is directly computed, effectively capturing the orientation and scale variability characteristic of SAR ship targets.

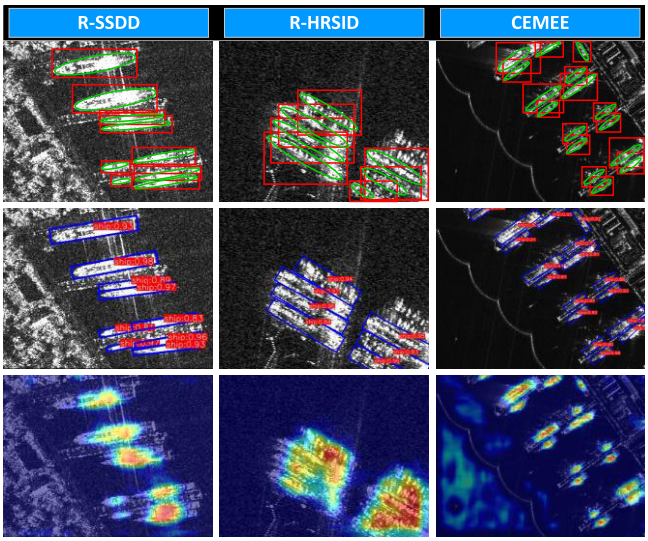


Fig. 1. Comparison of different annotation types for ship targets in complex SAR images, as well as visualization results of detection features by the proposed method.

The primary contributions of our method can be highlighted as follows.

1) The MSDFF-Net is proposed as an adaptable, arbitrary-oriented SAR ship detection method suitable for various scenarios. This approach effectively mitigates multiscale imbalance issues inherent in SAR ship detection and demonstrates robust performance in detection accuracy, parameter size, and computational efficiency.
2) Two feature-balancing modules, MSLK-Block and DFF-Block, were developed to address imbalances in feature scale and dimensionality. The former achieves balanced extraction of multiscale features by integrating large-kernel and partitioned heterogeneous convolution. The latter achieves balanced multiscale spatial–channel feature fusion through dynamic feature adaptation, ensuring more adaptive and discriminative feature integration for SAR ship detection.
3) To address the scale imbalance between regression and classification losses, a GPD loss function is introduced. By computing the Bhattacharyya distance between distributions, this approach refines ship bounding box regression, particularly for arbitrary-oriented multiscale ship targets in SAR images.
4) Extensive evaluation of the proposed model is conducted across three SAR ship detection datasets with varying resolutions, establishing standardized experiment-al benchmarks. Quantitative metrics and feature visualization results comprehensively confirm the capability of the proposed methodology.

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II. Related Works

A. Multiscale Feature Extraction Methods

In the SAR ship detection framework based on DL technology, the SAR image will be sent to the backbone network for feature extraction after preprocessing. The backbone network extracts multiscale target feature information from the input image and feeds the extracted features back to the subsequent detection module. Therefore, the effectiveness of feature extraction is directly related to the final detection performance of the model. To this end, a large number of scholars have improved the network feature extraction module based on the multiscale, small-scale, and complex scene characteristics of SAR ship detection, thereby improving the overall detection performance. In order to effectively extract the multiscale features of ship targets, Jia et al. [36] introduced the Swin transformer into the two-stage target detection model for ship target feature extraction. The constructed backbone network consists of four stages, each of which can extract effective feature information of different scales. Zhou et al. [37] used the pyramid vision transformer (PVT) to extract multiscale features of ships in SAR images. The proposed method can overcome the limitation of the insufficient receptive field of CNN through the self-attention mechanism and enhance the target feature extraction capability. Shen et al. [38] proposed ELLK-Net, which enhances the multiscale feature expression of anchor-free networks by adaptively selecting convolutional kernels. Deng et al. [39] redesigned a multiscale feature extraction backbone network consisting of multiple dense blocks. This method can effectively reduce the domain incompatibility problem caused by transfer pretraining. Shan et al. [40] proposed EMFENet, which uses a multifeature fusion structure to enhance the feature capture capability of the overall network structure. Collectively, these advancements underscore the importance of robust multiscale feature extraction tailored to the challenges of SAR imagery, forming the foundation for our proposed MSLK-Block that further addresses the multiscale imbalance in SAR ship detection.

B. Adaptive Feature Fusion Methods

The original image can be passed through the backbone network to obtain feature maps of different scales. Direct detection based on these feature maps often results in missed detection of small-scale targets. To solve this problem, feature pyramids are proposed to fuse feature maps of different layers. Cui et al. [41] proposed a dense attention pyramid network (DAPN), in which they embedded a convolutional block attention module (CBAM) between feature maps of different scales to achieve refinement and enhancement of the feature maps. Fu et al. [42] combined pixel-level attention and spatial attention to construct an attention balance pyramid to enhance the semantic characteristics of the target. Cui et al. [43] improved the CenterNet model by introducing a spatial shuffle group enhancement (SSE) attention module in the feature fusion part to enhance the target semantic feature expression. Ye et al. [44] proposed a channel space grouping attention mechanism and a feature transfer fusion module. Li et al. [45] used the polar coordinate Fourier transform to obtain the frequency-domain rotation characteristics of ship targets and fused the spatial- and frequency-domain features in multiple dimensions during feature fusion to further enhance the expression of ship target characteristics. In order to refine target features and suppress the influence of deformation, Zhang et al. [22] embedded a local and global context fusion module (LGCFM) at the front end of the feature pyramid to enhance the expression of target context information in feature fusion. These methods collectively emphasize adaptive fusion techniques; however, many are primarily designed for natural imagery and do not fully address the unique multiscale feature utilization imbalance in SAR images, where nonuniform scattering and clutter significantly impact detection performance. Our work builds upon these advancements by specifically tailoring an adaptive fusion strategy to mitigate SAR-specific challenges, ensuring robust target feature extraction in complex maritime environments.

C. Regression Loss Optimization Methods

In terms of scene distribution, ship targets often show dense layouts and docking, so many detection methods based on directed boxes have been proposed. In terms of target characteristics, ship targets have significant elliptical, multiscale, and small-scale characteristics. For this reason, many scholars have carried out loss optimization designs based on target characteristics. An et al. [46] proposed an improved directed box encoding scheme and used hard negative mining (HNM) technology when sampling positive and negative samples, which effectively alleviated the problem of unbalanced sample sampling in complex scenes. Liu et al. [29] introduced the evidence learning to capture cognitive uncertainty in the loss optimization process. They proposed the concept of sample bias learning and tried to use sample uncertainty to improve the imbalance of foreground and background distribution. Yue et al. [23] proposed a weakly supervised detection framework for SAR ship targets and combined it with the classic MaxIoU sample assignment method to achieve true and false sample allocation learning. In order to improve the constraints on the target bounding box during the iteration process, Zhang et al. [22] introduced centrality loss in the multitask loss function. To address the problem of poor detection of small targets, they further proposed a normalized centrality loss. Pu et al. [28] proposed a relational constraint loss based on the dependency between prediction confidence and IoU branches. While these approaches provide valuable insights, they primarily address generic detection challenges and do not fully capture the unique regression loss imbalance observed in SAR ship detection. In SAR images, the elliptical scattering patterns and pronounced scale variability led to inconsistent loss distributions between regression and classification tasks. To directly tackle these SAR-specific issues, our work introduces the GPD loss function. By computing the Bhattacharyya distance between the predicted and ground truth Gaussian distributions, our method effectively models the elliptical geometry of ship targets and harmonizes the optimization process between regression and classification. This tailored loss design significantly improves detection performance in complex maritime environments.

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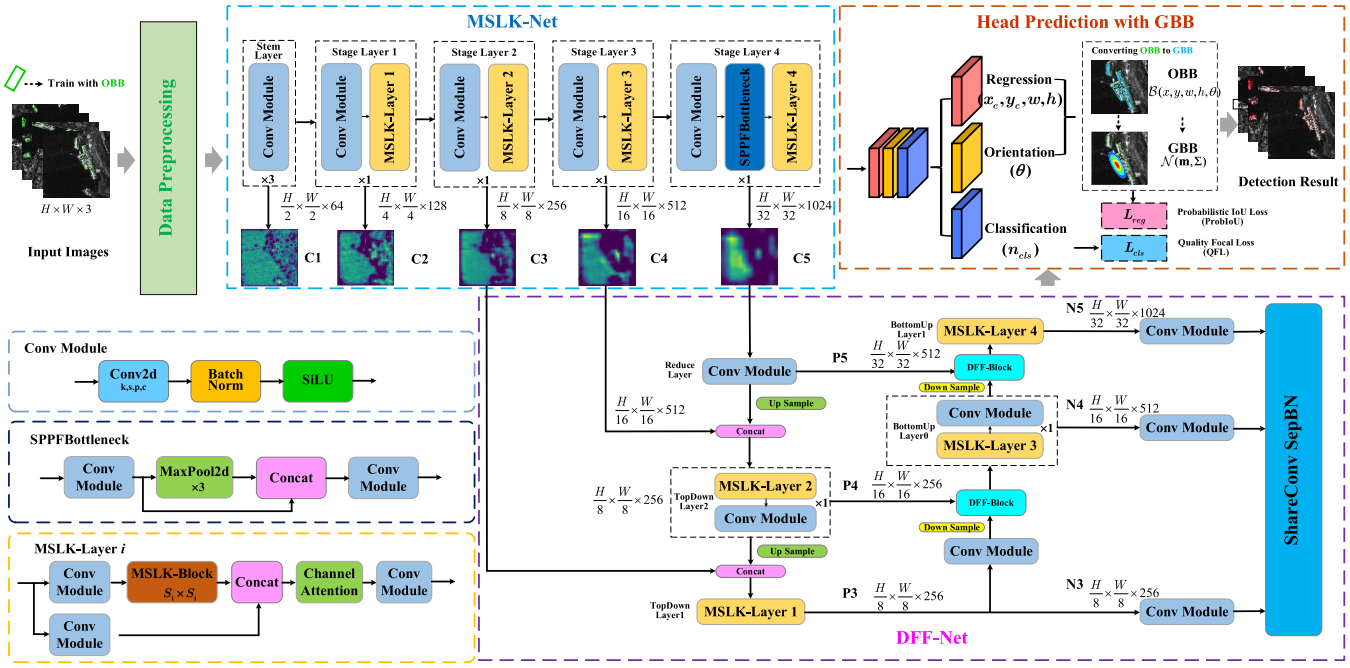


Fig. 2. Overall structure of the proposed MSDFF-Net, which primarily consists of three components: MSLK-Net, DFF-Net, and head prediction with GBB.

III. Methodology

This section first presents an outline of the overall architecture of our proposed approach. A detailed explanation of the specifically designed modules aimed at addressing the multiscale imbalance problem in arbitrary-oriented SAR ship detection is subsequently provided. For each module, in-depth theoretical analyses and visual demonstrations are included.

A. Overall Structure of MSDFF-Net

The proposed model's overall architecture is illustrated in Fig. 2. Our works are focused on the following three aspects.

1) Due to the diverse scales of ships and the strong background clutter in SAR images, capturing both fine-grained local details and long-range dependencies is crucial. To address this, we propose MSLK-Net, a multiscale large-kernel backbone that enhances feature extraction by leveraging large receptive fields while preserving detailed spatial information. This enables better multiscale representation of ship targets in complex scenes (see Section III-B for details).
2) SAR images often exhibit nonuniform target–background contrast, making it challenging to balance the contributions of spatial and channelwise features during fusion. To overcome this, we introduce a dynamic feature fusion (DFF) module, which adaptively integrates global context information to enhance spatial–channel feature interactions. This module is further incorporated into MSLK-Net, forming an enhanced feature pyramid structure named DFF-Net (see Section III-C for details).
3) Ship targets in SAR imagery typically exhibit elliptical scattering characteristics, and conventional bounding box regression may not effectively capture their geometric properties of ships at different scales. To tackle this, we reformulate the OBB representation using a GPD, inspired by the elliptical nature of ship targets.

In addition, we refine the loss function by integrating probabilistic IoU (ProbIoU) [47] to better balance classification and regression optimization. By leveraging the GPD distance, our method improves localization accuracy and mitigates the learning imbalance between regression and classification tasks (see Section III-D for details).

Specifically, assuming the original training dataset be denoted as $\mathcal{R}^{H\times W\times 3}=\{I_i,(x_i,y_i,w_i,h_i,\theta_i)\}_{i=1}^{N}$, where N is defined as the number of images used in the training process. Each image $I_i\in\mathcal{R}^{H\times W\times 3}$ has a corresponding OBB annotation represented as $B(x_i,y_i,w_i,h_i,\theta_i)$. The augmented dataset is obtained by $\mathcal{R}'=\{T(I_i)\}_{i=1}^{N'}$, T is the data augmentation operation, and $N'\geq N$ indicates the quantity of images in augmented dataset. Then, the input image I_i is fed into the MSLK-Net for feature extraction. There are a total of five layers of network structure in MSLK-Net, which will be depicted in Section III-B3. First, after three Conv modules $(\text{Conv}_1,\text{Conv}_2,\text{Conv}_3)$, the feature map $C_1\in\mathbb{R}^{(H/2)\times(W/2)\times 64}$ is generated. Then, C_1 is transmitted subsequent network to obtain $C_2\in\mathbb{R}^{(H/4)\times(W/4)\times 128}$, $C_3\in\mathbb{R}^{(H/8)\times(W/8)\times 256}$, $C_4\in\mathbb{R}^{(H/16)\times(W/16)\times 512}$, and $C_5\in\mathbb{R}^{(H/32)\times(W/32)\times 1024}$, respectively. Afterward, the feature maps C_3–C_5 are sent to the DFF-Net for feature fusion and enhancement. After the designed large kernel convolutional processing, the intermediate feature maps $P_3\in\mathbb{R}^{(H/8)\times(W/8)\times 256}$, $P_4\in\mathbb{R}^{(H/16)\times(W/16)\times 256}$, and $P_5\in\mathbb{R}^{(H/32)\times(W/32)\times 512}$ are generated. In the DFF-Net, the feature maps P_3–P_5 are dynamically fused by DFF-Block to obtain enhanced feature maps $N_3\in\mathbb{R}^{(H/8)\times(W/8)\times 256}$, $N_4\in\mathbb{R}^{(H/16)\times(W/16)\times 512}$, and $N_5\in\mathbb{R}^{(H/32)\times(W/32)\times 1024}$. In the inference stage, N_3, N_4, and N_5 are sent to the head prediction module for regression and classification through shared convolution. In the training stage, the OBB $B(x_i,y_i,w_i,h_i,\theta_i)$

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TABLE I
VARIABLE RELATIONSHIP BETWEEN INPUT AND OUTPUT OF MSDFF-NET

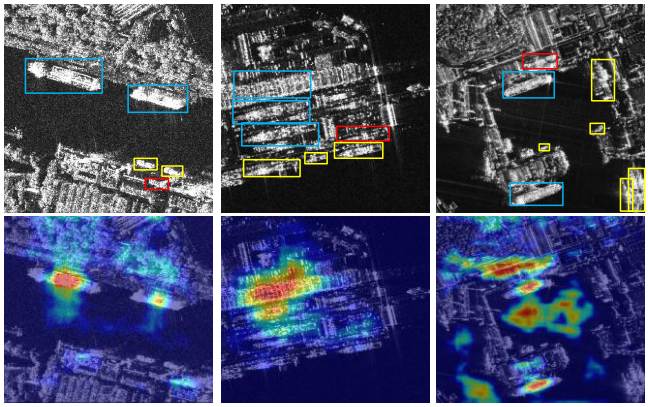
Task	Network	Module	Input	Output
Date Processing	-	Resize, Crop, Flip, etc Mosaic, MixUP, etc	$\mathcal{R}$	$\mathcal{R}'$
Feature Extraction	MSLK-Net	$\text{Conv}_1,\text{Conv}_2,\text{Conv}_3$	I_i	C_1
		MSLK-Layer	C_1	C_2,C_3,C_4,C_5
Feature Fusion	DFF-Net	MSLK-Layer	C_3,C_4,C_5	P_3,P_4,P_5
		DFF-Block	P_3,P_4,P_5	N_3,N_4,N_5
Head Prediction	Share Conv	Inference	N_3,N_4,N_5	$B(x_i,y_i,w_i,h_i,\theta_i)$
		Training	N_3,N_4,N_5	$L_{ProbIoU}$ L_{QFL}

is first converted into the Gaussian distribution $G(\mu,\sum)$ in head prediction. Finally, the GPD loss is used to calculate the regression loss, and the quality focal loss (QFL) [48] is used to calculate the classification loss. The training iteration is continued until the loss converges. The variable relationship between the input and output of the proposed MSDFF-Net is shown in Table I.

B. Multiscale Feature Extraction Balancing

1) Motivation: Effective and efficient extraction of multiscale features for SAR ships poses a considerable challenge due to the pronounced scale imbalance. This imbalance often results in the underrepresentation of smaller ships. Fig. 3 shows the multiscale characteristics obtained through the backbone network and the corresponding detection visualization results for ships in different datasets, based on the most advanced real-time detection model, real-time models for object detection (RTMDet) [49]. As shown in Fig. 3, strong scatterers in nearshore scenes can divert feature extraction to nontarget areas, leading to increased false alarms, while the extraction of small-scale target features remains weak, causing missed detections. Although previous works have addressed multiscale detection, they often incur heavy model complexity, lack cross-architecture adaptability, or sacrifice accuracy for lightweight design, issues that are particularly problematic in complex nearshore environments. This underscores the need for an efficient backbone network specifically tailored to the unique traits of SAR ship targets, enabling robust multiscale feature representation while mitigating false alarms and missed detections.

2) MSLK-Layer: In general object detection, large-kernel convolutions are widely adopted to enhance multiscale feature extraction. For instance, RTMDet employs a 5×5 large-kernel convolution within its backbone to boost feature extraction across its core modules. While this approach offers computational efficiency and improved extraction capabilities, its fixed kernel size limits adaptability to varying target scales, a critical concern in SAR imagery, where ship targets exhibit significant scale variations and are often obscured by background noise. To address these challenges, as shown in Fig. 4(b), we integrate large-kernel convolution with a multiscale partitioning mechanism to develop our novel MSLK-Layer. As illustrated



(a) (b) (c)

Fig. 3. Detection and feature visualization results of multiscale samples across different datasets using the RTMDet backbone network. In this figure, blue indicates correctly detected targets, yellow represents the missed targets, and red signifies incorrectly detected targets. (a) R-SSDD (b) R-HRSID (c) CEMEE.

in Fig. 4(a), the original CSPLayer in RTMDet applies a standard 3×3 convolution followed by a 5×5 depthwise large-kernel convolution, increasing computational complexity while processing features at a single scale. In contrast, our MSLK-Layer adopts a partitioned convolutional approach. In the lower branch, we utilize an inverted bottleneck (IBN) layer, inspired by ResNeXt [50], which expands the channel dimension using a 1×1 convolution and subsequently compresses it with another 1×1 convolution. This design not only reduces the parameter count and computational cost but also allows for the insertion of a large-kernel convolution between the expansion and compression stages, enhancing the nonlinear mapping capability necessary to capture diverse multiscale features. Overall, the MSLK-Layer is specifically designed to overcome the limitations of fixed large-kernel approaches by dynamically capturing multiscale ship features in SAR images, thereby significantly improving feature representation under challenging conditions.

First, we treat the input feature $F_{\text{input}} \in \mathbb{R}^{H\times W\times C}$ with a 1×1 convolution aimed at altering the channel count in extracted features to $3C/2$. This adjusted feature F'_{Input} is then passed through the MSLK-Block with a kernel size of $S_i \times S_i$ for further feature extraction, where

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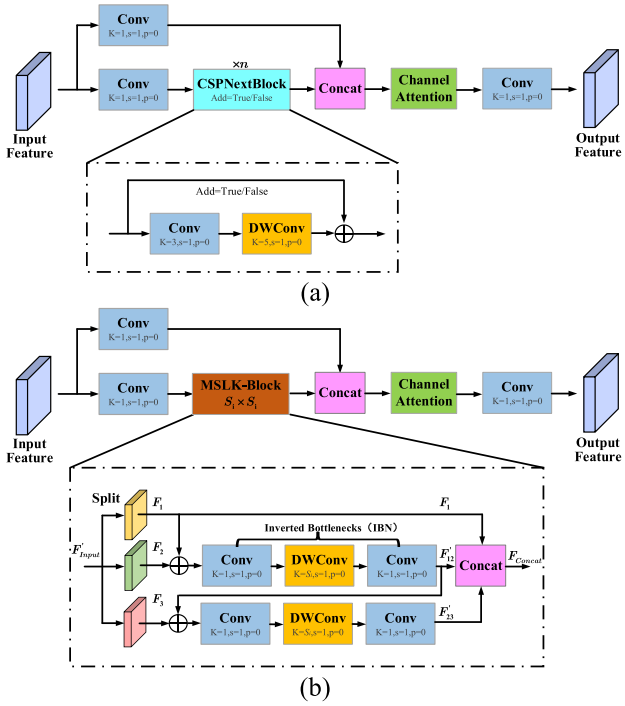


Fig. 4. Structure comparison between two basic feature extraction layers of backbone networks. (a) CSPLayer of RTMDet. (b) MSLK-Layer.

$S_i \times S_i$ is determined by the layer of the backbone network (details discussed in Section III-B2)

$$F'_{\text{Input}} = \text{Conv}_{1\times1}(F_{\text{Input}}) \in \mathbb{R}^{H\times W\times \frac{3C}{2}} \tag{1}$$

where $\text{Conv}_{1\times1}$ refers to the standard 1×1 convolution.

In the MSLK-Block, the F'_{Input} feature is split into j distinct groups, formulated as follows:

$$\{F_j\}_{j=1}^{3} = \text{Split}\left(F'_{\text{Input}}\right) \tag{2}$$

where $\{F_j\}_{j=1}^{3}$ represents the block features, j typically equals 1–3, and Split indicates a grouping operation.

To enhance intergroup feature relationships, we adopt an iterative approach that effectively extracts spatial dependencies. F_1 is left unaltered to retain the original feature information. F_2 is first fused with F_1 through an addition operation to obtain F_{12}. F_{12} is then passed through the $\text{IBN}_{S_i\times S_i}$ block for feature extraction. Since optical images typically have clearer, more defined edges and textures, the expansion convolution can maintain a moderate number of channels without introducing excessive computational complexity. However, SAR images require larger channel sizes in the expansion convolution to capture the more subtle, noisy features and preserve spatial context, especially for small- or low-contrast ship targets in cluttered environments. $\text{IBN}_{S_i\times S_i}$ consists of three parts: the first module is an expansion convolution that adjusts the feature channel count to obtain F_{12}^{Expanded}. F_{12}^{Expanded} is passed through large kernel depth-to-depth convolution and pointwise convolution, resulting in $F_{12}^{\text{Depthwise}}$ with an output channel of $\tau \cdot C/2$. Finally, $F_{12}^{\text{Depthwise}}$ is treated with a 1×1 convolution to adjusts the channel to $C/2$, resulting in F'_{12}. This is accomplished as follows:

$$F_{12} = F_1 \oplus F_2 \tag{3}$$

$$F_{12}^{\text{Expanded}} = \text{Conv}_{1\times1}(F_{12}) \in \mathbb{R}^{H\times W\times \tau\frac{3C}{2}} \tag{4}$$

$$F_{12}^{\text{Depthwise}} = \text{Conv}_{S_i\times S_i}\left(F_{12}^{\text{Expanded}}\right) \in \mathbb{R}^{H\times W\times \tau\frac{3C}{2}} \tag{5}$$

$$F'_{12} = \text{Conv}_{1\times1}\left(F_{12}^{\text{Depthwise}}\right) \in \mathbb{R}^{H\times W\times \frac{3C}{2}} \tag{6}$$

where τ is the expansion factor, the output channel count of the expansion layer is $\tau \cdot C$ and $\oplus$ denotes the feature elementwise addition.

For the third branch, F'_{12} is first fused with F_3 through addition, and then processed through $\text{IBN}_{S_i\times S_i}$ resulting in F'_{23}

$$F_{23} = F'_{12} \oplus F_3 \tag{7}$$

$$F_{23}^{\text{Expanded}} = \text{Conv}_{1\times1}(F_{23}) \in \mathbb{R}^{H\times W\times \tau\frac{3C}{2}} \tag{8}$$

$$F_{23}^{\text{Depthwise}} = \text{Conv}_{S_i\times S_i}\left(F_{23}^{\text{Expanded}}\right) \in \mathbb{R}^{H\times W\times \tau\frac{3C}{2}} \tag{9}$$

$$F'_{23} = \text{Conv}_{1\times1}\left(F_{23}^{\text{Depthwise}}\right) \in \mathbb{R}^{H\times W\times \frac{3C}{2}}. \tag{10}$$

After that, F_1, F'_{12}, and F'_{23} are combined along the channel axis and fed into a channel attention mechanism for dynamic adjustment at the channel level

$$F_{\text{Concat}} = \text{Concat}\left(F_1, F'_{12}, F'_{23}\right) \in \mathbb{R}^{H\times W\times \frac{C}{2}} \tag{11}$$

$$F'_{\text{Concat}} = \text{Concat}\left(F'_{\text{Input}}, F_{\text{Concat}}\right) \in \mathbb{R}^{H\times W\times C} \tag{12}$$

$$F_{\text{Squeeze}} = \text{AdaptiveAvg}\left(F'_{\text{Concat}}\right) \in \mathbb{R}^{1\times1\times C} \tag{13}$$

$$F_{\text{Excitation}} = \sigma(\text{FC}_2(\text{ReLU}(\text{FC}_1(F_{\text{Squeeze}})))) \tag{14}$$

$$F_{\text{CA}} = F'_{\text{Concat}} \otimes F_{\text{Excitation}} \tag{15}$$

where $\text{Concat}(\cdot)$ represents the feature concatenation along the channel dimension. $\text{AdaptiveAvg}(\cdot)$ denotes the adaptive average pooling module, FC is indicated as fully connected layer, σ denotes a classical HardSigmoid function, ReLU represents a linear activation function, and $\otimes$ indicates an elementwise multiplication.

Finally, F_{CA} is passed through a standard 1×1 convolution to produce the output feature F_{output}, with dimensions $\mathbb{R}^{H\times W\times C}$, formulated as follows:

$$F_{\text{Output}} = \text{Conv}_{1\times1}(F_{\text{CA}}) \in \mathbb{R}^{H\times W\times C}. \tag{16}$$

The design of the MSLK-Layer offers two significant advantages. First, by partitioning features, it captures interblock dependencies within complex scenes, effectively preserving multilevel feature information and achieving efficient multiscale feature fusion. Second, the IBN-based feature extraction approach enhances the extraction of key features while maintaining the computational efficiency of large-kernel convolutions. Unlike conventional large kernel convolutional approaches such as those used in LSKNet [51], which are primarily designed for natural images and mainly focus on capturing long-range dependencies. Our MSLK-Layer is specifically tailored for the unique challenges of SAR imagery, which can effectively capture the diverse scattering characteristics and scale variations of ship targets in SAR images, leading to a more balanced and robust feature representation.

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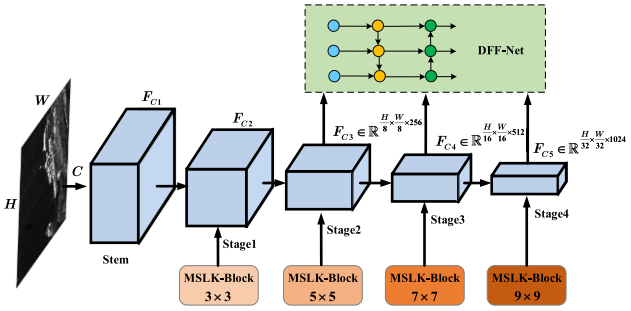


Fig. 5. Structural diagram of MSLK-Net.

3) MSLK-Net: By integrating the proposed MSLK-Block, a novel backbone network, termed MSLK-Net, was developed, as illustrated in Fig. 5. In designing the core feature extraction module, we observed that the receptive field size critically impacts detection performance, particularly for ship targets with significant scale variability. Traditional detection models rely on a fixed convolutional kernel size throughout the backbone, which poses two challenges: for small-scale ships, the receptive field may be too large, leading to insufficient fine-grained feature extraction; for large-scale ships, the receptive field may include excessive background information. These limitations hinder overall model performance. To overcome this, MSLK-Net adaptively adjusts the large-kernel convolutional size within each MSLK-Block according to the scale of the feature maps. Specifically, multiple large depth kernels at various scales are applied to capture the ship's multiscale features. Rather than directly merging different large-depth kernels, MSLK-Net employs heterogeneous depthwise convolutions, dynamically adjusting the kernel size k_i for each MSLK-Block. The sequence of kernel sizes k is set to 3, 5, 7, and 9. At shallow layers, a 3 × 3 convolution focuses on fine-grained details, while deeper layers progressively use larger kernels to capture broader contextual information. This adaptive mechanism expands the receptive field at higher levels, improving spatial understanding and enhancing detection and localization precision. Experimental results confirm that our approach not only improves convolutional efficiency and reduces computational costs, but also enriches high-level, low-resolution feature maps with more nuanced multiscale information, ultimately leading to superior performance in SAR ship detection. In optical images, large kernels capture global context and semantic patterns, with relatively uniform kernel sizes due to the clearer, less noisy details. In contrast, SAR images present challenges, such as speckle noise, clutter, and nonuniform scattering. To address these, MSLK-Net uses larger kernels at certain stages to better capture ships at varying scales and orientations, especially for small-, low-contrast targets in noisy SAR environments.

C. Spatial–Channel Feature Fusion Balancing

Most existing SAR ship detection models adopt a static, uniform fusion strategy in feature integration, where identical fusion weights are applied across all input images. However, this approach fails to adapt to the inherent variations in scale, scattering characteristics, and local target changes unique to SAR images, leading to suboptimal feature utilization.

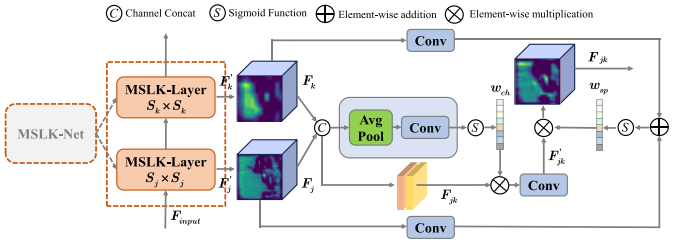


Fig. 6. DFF-Block structure combined with the spatial adaptation and channel optimization.

Moreover, conventional methods often overemphasize channelwise correlations while underutilizing spatial structural information, limiting their effectiveness in distinguishing ships from complex backgrounds. In addition, the lack of cross-scale correlation modeling results in insufficient extraction of discriminative fusion features. Ship targets in SAR images exhibit nonuniform scattering distributions, target–background contrast variations, and complex interference from sea clutter and speckle noise, making it challenging for traditional feature fusion methods to achieve optimal feature selection. To mitigate this, we propose the DFF module, designed to adaptively balance spatial and channelwise feature contributions. Unlike fixed-weight fusion strategies, DFF dynamically adjusts fusion weights according to the global context information, ensuring that essential spatial and channel-level features are appropriately emphasized. This mechanism allows the network to better capture discriminative structural and scattering features of ships, leading to improved feature integration across different scales. By embedding DFF into the backbone network, we establish an adaptive multiscale feature pyramid, effectively addressing the imbalance in multiscale feature utilization, and enhancing SAR ship detection performance in complex maritime environments. The specific implementation is detailed as follows.

1) DFF-Block: The proposed feature fusion method leverages the global spatial scale correlations between higher and lower level feature maps within the backbone network, dynamically optimizing the channel features to generate more discriminative multiscale pyramid features. The configuration of our DFF-Block is presented in Fig. 6 and mainly comprises two components: spatial feature adaptation and channel feature optimization.

Specifically, assume that the input feature is $F_{\text{input}} \in \mathbb{R}^{H\times W\times C}$, from which multiscale features are obtained through the backbone network. Taking the kth and jth layers as examples, $F_k' \in \mathbb{R}^{H_k\times W_k\times C_k}$ and $F_j' \in \mathbb{R}^{H_j\times W_j\times C_j}$ represent the higher and lower level scale features extracted by the backbone network, respectively. Due to differences in spatial and channel scales between the features, it is necessary to adjust the spatial and channel scales of F_k' and F_j' to obtain the input features F_k and F_j before feature fusion. The adjusted features are then concatenated to produce the fused feature $F_{jk} \in \mathbb{R}^{H_{jk}\times W_{jk}\times C_{jk}}$, formulated as follows:

$$F_j = \text{adjust}\left(F_j'\right),\ F_j' \in \mathbb{R}^{H_j\times W_j\times C_j} \tag{17}$$

$$F_k = \text{adjust}\left(F_k'\right),\ F_k' \in \mathbb{R}^{H_k\times W_k\times C_k} \tag{18}$$

$$F_{jk} = \text{Concat}(F_j, F_k) \in \mathbb{R}^{H_{jk}\times W_{jk}\times C_{jk}}. \tag{19}$$

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In the conventional static fusion, concatenation is used to obtain the fused feature. F_{jk}. In our dynamic fusion framework, F_{jk} serves as the base fused feature is refined at both spatial and channel levels. For global channel optimization, we apply global average pooling to derive a spatial attention vector $F_{\text{Avg}} \in \mathbb{R}^{1\times1\times C_{jk}}$, pass it through a $1 \times 1 \times 1$ convolution to obtain a convolutional vector $F_{\text{Conv}} \in \mathbb{R}^{1\times1\times C_{jk}}$, and then normalize it with a sigmoid function to generate the final channel attention weights $w_{\text{ch}} \in \mathbb{R}^{1\times1\times C_{jk}}$

$$F_{\text{Avg}}(c) = \frac{1}{H \times W} \sum_{h=1}^{H_{jk}} \sum_{w=1}^{W_{jk}} F_{jk}(h, w, c) \quad \text{where } c = 1, 2, \ldots, C_{jk} \tag{20}$$

$$F_{\text{Conv}} = \text{Conv}_{1\times1\times1}(F_{\text{Avg}}) \tag{21}$$

$$w_{\text{ch}} = \sigma(F_{\text{Conv}}) \tag{22}$$

where $F_{jk}(h, w, c)$ denotes the feature value of F_{jk} at channel c and spatial position (h, w), $F_{\text{Avg}}(c)$ represents the feature value F_{jk} after global average pooling, and $\text{Conv}_{1\times1\times1}$ denotes the $1 \times 1 \times 1$ convolutional operation.

During the global spatial feature optimization stage, the two input feature branches are first processed with the application of a 1×1 convolution to adjust the corresponding weights, $w_{\text{sp}}^{k} \in \mathbb{R}^{H_{jk}\times W_{jk}\times1}$ and $w_{\text{sp}}^{j} \in \mathbb{R}^{H_{jk}\times W_{jk}\times1}$. The two weights are then summed elementwise to produce the global spatial attention weight $w_{\text{sp}} \in \mathbb{R}^{H_{jk}\times W_{jk}\times1}$

$$w_{\text{sp}}^{j} = \text{Conv}_{1\times1\times1}(F_j) \in \mathbb{R}^{H_{jk}\times W_{jk}\times1} \tag{23}$$

$$w_{\text{sp}}^{k} = \text{Conv}_{1\times1\times1}(F_k) \in \mathbb{R}^{H_{jk}\times W_{jk}\times1} \tag{24}$$

$$w_{\text{sp}} = \sigma\left(w_{\text{sp}}^{j} \oplus w_{\text{sp}}^{k}\right) \tag{25}$$

where $\oplus$ indicates elementwise addition. Finally, the global channel attention weight w_{ch} and the global spatial attention weight w_{sp} are used to optimize the base fused feature F_{jk}. The weight w_{ch} enhances the selection of important features at the channel level, aiding in the retention of more critical channel-level features. Meanwhile, the weight w_{sp} reinforces the spatial dependencies between local feature mappings, facilitating the alignment of spatial characteristics within the features.

2) DFF-Net: A novel feature fusion pyramid, DFF-Net, was constructed by integrating the DFF module. The configuration of our pyramid is represented in Fig. 7. The feature fusion component employs the same MSLK-Layer as our main structure. Throughout the feature fusion process across various levels, DFF is performed using the DFF-Block. In DFF-Net, the input image $I_i \in \mathcal{R}^{H\times W\times3}$ is processed by MSLK-Net to gather different levels of features $\{C_1, C_2, C_3, C_4, C_5\}$. To balance the computational efficiency with detection performance, the top three layers of features $\{C_3, C_4, C_5\}$ are selected for feature fusion. The intermediate layer feature $\{P_3, P_4, P_5\}$ are obtained by downsampling and channel concatenation. This feature is then subjected to DFF using our designed DFF module, resulting in the optimized feature $\{N_3, N_4, N_5\}$. Finally, $\{N_3, N_4, N_5\}$ are fed into the detection head for result prediction. In DFF-Net, a dynamic fusion mechanism balances channel and spatial features across multiscale pyramids, mitigating the feature utilization imbalance in SAR ship detection

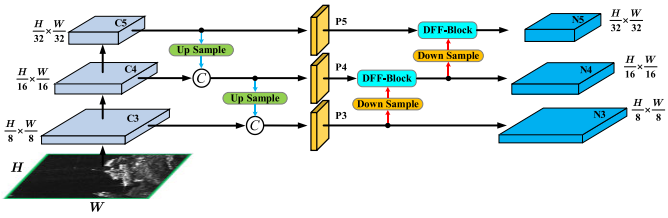


Fig. 7. Overview of DFF-Net: integrating multiscale features and dynamic fusion.

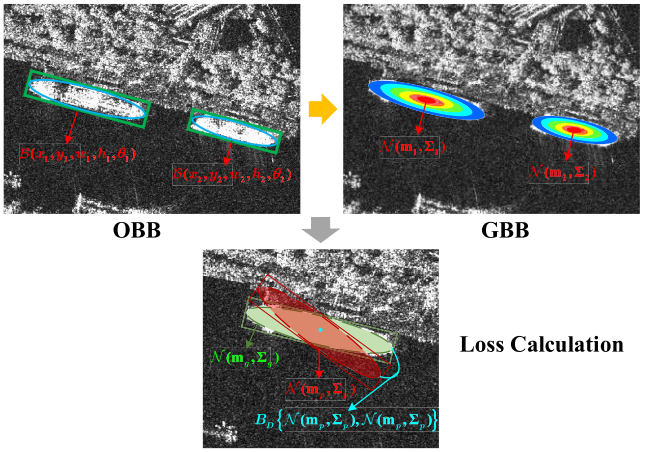


Fig. 8. Diagram illustrating the transformation of ship representation and loss calculation.

with minimal added complexity and enhancing overall accuracy. The dynamic fusion primarily focuses on refining the features across different spatial levels and channels in optical images. However, increased dynamic attention across spatial and channel dimensions to handle the SAR-specific issue of clutter and weak target boundaries. Higher focus on spatial locality in SAR data due to the significant impact of noise and scattering, making it crucial to better highlight local features.

D. GPD Loss Function

Due to SAR's imaging mechanism, strong scattering points (e.g., from ship masts or edges) may shift the apparent centroid of a ship, causing the regression target to deviate from the actual object extent. This effect is particularly pronounced for high-resolution SAR images where subpixel localization errors can significantly impact detection performance. Meanwhile, standard IoU-based losses tend to favor large objects since their relative position errors are lower compared with smaller ships. This imbalance leads to suboptimal regression accuracy for small-scale targets. To address these challenges, we introduce a ProbIoU loss, which models the bounding box as a 2-D GPD. The overall multitask loss of the proposed model comprises two components: $\mathcal{L}_{\text{reg}}$ and $\mathcal{L}_{\text{cls}}$. $\mathcal{L}_{\text{reg}}$ represents the boundary regression loss, while $\mathcal{L}_{\text{cls}}$ denotes the classification loss. In our model, $\mathcal{L}_{\text{reg}}$ loss has been optimized by incorporating the distance constraint between Gaussian distributions to represent the boundary regression loss. Fig. 8 depicts the process of transforming the bounding box representation method along with the corresponding loss calculation. The details of the loss $\mathcal{L}_{\text{reg}}$ and $\mathcal{L}_{\text{cls}}$ are provided below. First,

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the representation OBB $\mathcal{B}(x, y, w, h, \theta)$ is converted a specific 2-D GPD $\mathcal{N}(\mathbf{m}, \Sigma)$. Here, (x, y) denotes the coordinates of the center representing an OBB ground truth, w and h indicate the corresponding length and width, and θ signifies the rotation angle of OBB.

The $\mathcal{N}(\mathbf{m}, \Sigma)$ is characterized by a mean vector $\mathbf{m}$ and a covariance matrix Σ. $\mathbf{m}$ corresponds to a center of OBB and is calculated using the following formula:

$$\mathbf{m} = \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}. \tag{26}$$

The covariance matrix Σ encapsulates the spread and orientation of the distribution. For a bounding box with no rotation ($\theta = 0$), the covariance matrix is initially represented as follows:

$$\Sigma_0 = \begin{bmatrix} \left(\frac{w}{2}\right)^2 & 0 \\ 0 & \left(\frac{h}{2}\right)^2 \end{bmatrix}. \tag{27}$$

Here, Σ_0 represents the unrotated covariance, where $(w/2)$ and $(h/2)$ denote the standard deviations along the major and minor axes, respectively. To account for the rotation angle θ, we apply a rotation matrix $R(\theta)$ to transform the covariance matrix

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}. \tag{28}$$

The rotated covariance matrix Σ is then obtained as follows:

$$\Sigma = R(\theta)\Sigma_0 R(\theta)^{\mathrm{T}}. \tag{29}$$

Substituting the values, the covariance matrix Σ is given by

$$\Sigma = \begin{bmatrix} \frac{w^2}{4}\cos^2\theta + \frac{h^2}{4}\sin^2\theta & \frac{1}{4}(w^2 - h^2)\cos\theta\sin\theta \\ \frac{1}{4}(w^2 - h^2)\cos\theta\sin\theta & \frac{w^2}{4}\sin^2\theta + \frac{h^2}{4}\cos^2\theta \end{bmatrix}. \tag{30}$$

Assuming that $\mathcal{B}(x_p, y_p, w_p, h_p, \theta_p)$ represents the OBB predicted parameters regressed by the network and $\mathcal{B}(x_g, y_g, w_g, h_g, \theta_g)$ represents the OBB labels, the corresponding GBB distributions are $\mathcal{N}(\mathbf{m}_p, \Sigma_p)$ and $\mathcal{N}(\mathbf{m}_g, \Sigma_g)$, respectively. The Bhattacharyya coefficient B_c and Bhattacharyya distance B_d between the two Gaussian distributions are calculated as follows:

$$\begin{aligned} &B_c\{\mathcal{N}(\mathbf{m}_p, \Sigma_p); \mathcal{N}(\mathbf{m}_g, \Sigma_g)\} \\ &= \frac{1}{2\pi\sqrt{\sqrt{\det(\Sigma_p)\cdot\det(\Sigma_g)}}} \\ &\quad \times \exp\left(\left(-\frac{1}{8}(\mathbf{m}_p - \mathbf{m}_g)^{\mathrm{T}}\left(\frac{\Sigma_p + \Sigma_g}{2}\right)^{-1}(\mathbf{m}_p - \mathbf{m}_g)\right)\right) \end{aligned} \tag{31}$$

$$\begin{aligned} &B_d\{\mathcal{N}(\mathbf{m}_p, \Sigma_p); \mathcal{N}(\mathbf{m}_g, \Sigma_g)\} \\ &= \frac{1}{8}(\mathbf{m}_p - \mathbf{m}_g)^{\mathrm{T}}\left(\frac{\Sigma_p + \Sigma_g}{2}\right)^{-1}(\mathbf{m}_p - \mathbf{m}_g) \\ &\quad + \frac{1}{2}\ln\left(\frac{\det\left(\frac{\Sigma_p + \Sigma_g}{2}\right)}{\sqrt{\det(\Sigma_p)\cdot\det(\Sigma_g)}}\right) \end{aligned} \tag{32}$$

where $\det(\cdot)$ refers to the determinant function, which is a scalar value that can be computed from a square matrix.

The Bhattacharyya distance comprehensively accounts for both the central location and shape of the distributions, providing a more accurate measure of the similarity between two 2-D Gaussian distributions. During the training phase, B_d cannot be directly used for network regression and must be further transformed into the Hellinger distance H_d

$$H_d = \sqrt{1 - B_d\{\mathcal{N}(\mathbf{m}_p, \Sigma_p), \mathcal{N}(\mathbf{m}_g, \Sigma_g)\}}. \tag{33}$$

Then, based on the ProbIoU loss calculation standard [47], the distribution metric regression loss is obtained as follows:

$$\mathcal{L}_{\mathrm{reg1}} = H_d \in [0, 1] \tag{34}$$

$$\mathcal{L}_{\mathrm{reg2}} = -\ln\left(1 - \mathcal{L}_{\mathrm{reg1}}^2\right) \in [0, \infty]. \tag{35}$$

Here, $\mathcal{L}_{\mathrm{reg1}}$ and $\mathcal{L}_{\mathrm{reg2}}$ represent the regression loss values at different stages of training, respectively.

In this article, the QFL [48] is used to calculate the classification loss $\mathcal{L}_{\mathrm{cls}}$

$$\begin{aligned} \mathcal{L}_{\mathrm{cls}} = -\frac{1}{N_{\mathrm{pos}}}\sum_{i=1}^{N}&\left(y_i \log(\widehat{p}_i) + (1 - y_i)\log(1 - \widehat{p}_i)\right) \\ &\times \left|y_i - \widehat{p}_i\right|^{\gamma} \end{aligned} \tag{36}$$

where N_{pos} stands for the quantity of positive instances, y_i denotes the IoU value between the forecast bounding box and annotated box, $\widehat{p}_i$ refers to the probability value predicted by the corresponding model, and γ is the decay factor for adjusting the difficulty of samples (typically set to 2.0), which controls the weight of easy and hard samples.

Finally, the overall training loss for our model is calculated as follows:

$$\mathcal{L}_{\mathrm{all}} = \mathcal{L}_{\mathrm{cls}} + \lambda\mathcal{L}_{\mathrm{reg1}} + \beta\mathcal{L}_{\mathrm{reg2}} \tag{37}$$

where $\mathcal{L}_{\mathrm{all}}$ represents the total loss value and λ and β are the training adjustment factors that dynamically adjust the regression loss component based on the number of training epochs.

IV. Experiments

This section first introduces the three datasets utilized in our experiments, the evaluation criteria, and the corresponding experimental parameter settings. After that, we give detailed ablation experiment comparison results, as well as corresponding visualization, quantitative experimental results, and P–R curve analysis.

A. Datasets

1) R-SSDD: SSDD [33] is the inaugural dataset explicitly created for detecting ship targets in SAR imagery, featuring both HBB and OBB annotation formats. This dataset comprises images from three radar satellites: RadarSat-2, TerraSAR-X, and Sentinel-1, which offers resolutions that

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vary between 1 and 15 m. Ship targets are distributed across offshore and inshore scenes. The original dataset includes 1160 images and 2456 ship instances, with image sizes ranging from 190 to 688 pixels. To standardize the input image dimensions, the original dataset was processed to resize all images to 512 × 512 pixels, which yields in altogether 1247 images (872 images are assigned to training, while 375 are used for testing). In addition, the annotations of this dataset were revised based on expert experience, yielding the R-SSDD used in our experiments. The R-SSDD contains a total of 2853 ship instances (1975 for training and 878 for testing).

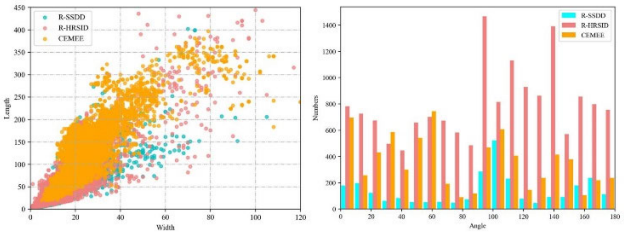
2) R-HRSID: HRSID [34] represents the initial dataset specifically designed for high-resolution SAR ship instance segmentation, featuring HBB and mask annotation formats. This dataset is composed of images from satellites, including Sentinel-1B, TerraSAR-X, and TanDEM-X, with resolutions varying from 0.5 to 3 m. Ship targets are distributed across offshore and inshore scenes. The original dataset consists of 5604 images and 16 951 ship instances, with each image sized at 800 × 800 pixels. The mask annotations in this dataset were converted to OBB annotations, and then further refined based on expert experience, resulting in the R-HRSID used in our experiments. The R-HRSID contains a total of 16 956 ship instances (11 038 for training and 5918 for testing).

3) CEMEE: CEMEE [35] dataset is an open dataset from the 8th CEMEE Cup Electromagnetic Environment Mathematical Modeling Detection and Recognition Algorithm Challenge, featuring an OBB annotation format. This dataset consists entirely of images from the Gaofen-3 radar satellite, with a resolution of 1 m. Ship targets are distributed across offshore and inshore scenes. After manual correction, the dataset contains 1500 images of size 1024 × 1024 pixels (1000 for training and 500 for testing), with a total of 7254 ship instances (4822 images are designated for training, while 2432 images are allocated for testing). Also, original CEMEE dataset has seven categories (S1–S7); for our experiments, all categories were merged into a single "ship" class to focus on detection performance.

The relevant information for these three datasets is summarized in Table II. Fig. 9 presents the length, width, and angle distribution statistics for the three datasets. From Fig. 9, it can be clearly observed that the three datasets exhibit strong similarity in terms of scale, while differences exist in the angle distribution in terms of quantity. However, the overall trend shows a certain degree of similarity. To facilitate algorithm validation by researchers, all experimental data used in this article have been uploaded to the website: https://github.com/SZZ-SXM/MSDFF-Net. Any corrections or inquiries are welcome.

TABLE II
COMPARISON OF THREE DATASETS USED IN OUR EXPERIMENT

Dataset	Images (num)	Ship (num)	Size (pixels)	Resolution (m)	Satellites
R-SSDD	1247	2853	512	1-15	RadarSat-2 TerraSAR-X Sentinel-1
R-HRSID	5604	16956	800	0.5-3	Sentinel-1B TerraSAR-X TanDEM-X
CEMEE	1500	7254	1024	1	Gaofen-3



(a) (b)

Fig. 9. Distribution of experimental datasets. (a) Length and width distribution of OBB annotations. (b) Angle distribution of OBB annotations.

B. Evaluation Criteria

The detection accuracy of different methods is first assessed through metrics precision (P) and recall (R). P indicates the proportion of true positives (TPs) among all samples classified as positive. The computation is given by

$$P = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{38}$$

where TPs indicate the accurately detected ships, while false positives (FPs) signify the count of incorrectly detected ships. R represents the proportion of TP samples accurately identified by the model to all actual positive samples, and it is expressed as Follows:

$$R = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{39}$$

where false negatives (FNs) denotes the count of ships that were missed. The $F1$-score, which serves as the balanced detection metric of P and R, is utilized to provide a comprehensive assessment of the model's ability, which can be calculated as follows:

$$F1\text{-score} = 2 \times \frac{P \times R}{P + R}. \tag{40}$$

Meanwhile, the mean average precision (mAP) is defined as follows:

$$\text{mAP} = \int_0^1 P(R) dR \tag{41}$$

where $P(R)$ is a curve of precision–recall at an IoU threshold of 0.5. In addition, Parameters (Params), floating point operations (FLOPs), and frames per second (FPS) are employed to provide a comprehensive analysis of the complexity and computational ability of different models.

C. Experimental Setting

All experiments were conducted using PyTorch 1.12.1, CUDA 11.6, and CuDNN 8.3.2, with an Intel[1] Xeon[1] Gold 6139 CPU operating at 2.30 GHz and a single NVIDIA GeForce RTX 3090 GPU. Our PC operation system is Ubuntu 18.04. All comparative experiments, including both anchor-free, as well as one-stage and two-stage detection

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TABLE III
EFFECT OF MSLK-NET IN DIFFERENT SCENES OF R-SSDD (%)

Scene	Method	P	R	mAP	F1-Score
Inshore	Baseline	84.32	83.61	80.22	83.96
	+ MSLK-Net	85.21	84.95	81.47	85.08
Offshore	Baseline	94.75	95.94	90.22	95.34
	+ MSLK-Net	95.47	96.62	91.03	96.04

algorithms, were conducted using the basic settings of the MMRotate [52]. The backbone networks were initialized with pretrained weights from ImageNet to accelerate model convergence. Following the work of Zhou et al. [37], the models were trained using either the AdamW or stochastic gradient descent (SGD) optimizer. The initial setting of the model learning rate for the AdamW was 0.0001, accompanied by a regularization term of 0.0005. For SGD, its initial learning rates were assigned a value of 0.001 or 0.005, with a momentum parameter of 0.9 and a regularization coefficient rate of 0.0001. In our experiments, a MultiStepLR method is used to automatically adjust the learning rate. The input size of detection algorithms was adjusted according to the scale of the input dataset.

D. Ablation Experiments and Analysis

1) Effect of MSLK-Net: The RTMDet-OBB [49] is established as the foundational baseline model for ablation studies. Based on the baseline algorithm, our proposed module schemes were progressively adopted, and comparative analyses were undertaken to emphasize the contribution of every component.

Table III presents the quantitative experimental results of the proposed backbone model compared with the baseline model on various scenarios in the R-SSDD dataset. Visual inspection reveals that our MSLK-Net significantly improves detection performance in both nearshore and offshore scenarios. For nearshore scenes, the feature extraction with MSLK-Net achieves a detection precision of 85.21% and a recall of 84.95%, with mAP and $F1$-score reaching 81.47% and 85.08%, respectively. Compared with the baseline model, these metrics show an improvement of 0.89%, 1.34%, 1.25%, and 1.12%, respectively. For offshore scenes, the proposed feature extraction method improves the four metrics by 0.72%, 0.68%, 0.81%, and 0.70% compared with the baseline model, reaching 95.47%, 96.62%, 91.03%, and 96.04%, respectively. The results in Table III further validate that the proposed MSLK-Net enhances the model's proficiency in extracting features from ships of varying scales, thereby improving target detection performance.

2) Effect of DFF-Net: In addition to optimizing the feature extraction module, the proposed algorithm also improves the feature fusion process. By incorporating a dynamic fusion mechanism, features extracted from different layers of the backbone model are dynamically integrated using our DFF-Net, effectively enhancing the semantic association between global and local scene information. To assess the efficacy of our feature fusion approach, we performed a visual analysis of the fused features using Eigen-CAM [53]. Eigen-CAM provided valuable insights into how the network attends to different regions of the image, especially those related to ship targets. This made it an ideal tool for validating the feasibility and effectiveness of DFF-Net in SAR ship detection, without introducing unnecessary computational complexity. In combination with quantitative results, feature visualizations, and detection performance evaluations, Eigen-CAM helped us demonstrate the effectiveness of the feature fusion strategy within DFF-Net. Fig. 10 shows the Eigen-CAM visualizations of multiscale fused features on R-SSDD images, comparing results with and without DFF-Net based on the baseline model. The red regions indicate areas where features have a high activation value or importance in predicting target class and locations. The first column shows different R-SSDD image samples with ground-truth annotations, the second column presents visualization outcomes from the comparison model, and the results after applying DFF-Net for feature fusion are displayed in the third column. Compared with the baseline model, DFF-Net demonstrates better spatial-scale feature fusion. This is especially evident in nearshore scenes, where the fused features concentrate more on target regions and reduce attention to land interference, as clearly shown in Fig. 10(a), (e), and (f). This demonstrates that the proposed DFF-Net model enhances the focus on targets between global and local semantics, improving the detection performance in complex scenes. In addition, Fig. 10(b)–(d) illustrates that the proposed method significantly enhances the model's ability to distinguish between adjacent targets. Fig. 10(b) shows more pronounced activation values compared with the baseline model, while Fig. 10(d) demonstrates that DFF-Net also strengthens the model response to individual offshore ships. This effectively explains the performance improvement in the offshore scene. Furthermore, Table IV presents the quantitative results of the overall ablation analysis for our three introduced modules. A comparison of the second and third columns in Table IV clearly demonstrates that the use of DFF-Net for feature fusion improves precision and recall by 0.91% and 0.97%, respectively, across all scenes in the R-SSDD dataset. The comprehensive detection performance, in terms of mAP and $F1$-score, improves by 0.53% and 0.94%, reaching 91.12% and 93.81%, respectively. The quantitative results in Table IV, together with the feature visualization in Fig. 10, confirm that the proposed feature fusion method effectively suppresses land interference in complex scenes, enhances scene-level semantic information, and improves model detection performance.

3) Effect of GPD Loss: To better evaluate the effectiveness of different prediction methods on angle estimation, we use the (MAE) to verify the impact of the proposed metric on angle prediction. The MAE is calculated as follows:

$$\mathrm{MAE} = \frac{1}{N}\sum_{i=1}^{N}\left|\theta_{\mathrm{pred},i} - \theta_{\mathrm{gt},i}\right| \tag{42}$$

where $\theta_{\mathrm{pred},i}$ represents an angle corresponding to ith predicted bounding box, $\theta_{\mathrm{gt},i}$ indicates an angle corresponding to ith ground truth bounding box, and N signifies a total number of detection samples.

The angular error calculation must account for periodicity. For instance, in the case of long edge definition [31], 180° and 0° should be regarded as close angles. Therefore, the

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TABLE IV
PERFORMANCE OF ABLATION EXPERIMENTS ON THE R-SSDD DATASET ACROSS VARIOUS SCENES (%)

MSLK-Net	DFF-Net	GPD loss	P	R	mAP	F1-Score	FPS
×	×	×	91.97	92.60	89.98	92.28	139.28
✓	×	×	92.54(+0.57)	93.21(+0.61)	90.59(+0.61)	92.87(+0.59)	137.47
✓	✓	×	93.45(+0.91)	94.18(+0.97)	91.12(+0.53)	93.81(+0.94)	134.22
✓	✓	✓	93.95(+0.50)	94.72(+0.54)	91.55(+0.43)	94.33(+0.52)	135.79

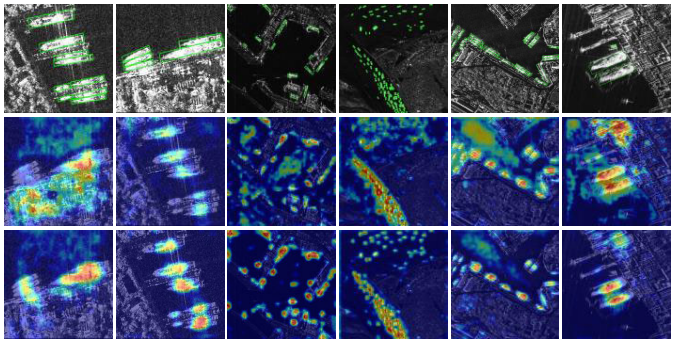


Fig. 10. Visualization comparison of features over several datasets. Initial SAR imagery alongside the ground truth is displayed in the first row. The corresponding CAM visualization results of baseline is presented in the second row. The CAM visualization results of baseline with DFF-Net are shown in third row. (a) and (b) Samples of R-SSDD. (c) and (d) Samples of R-HRSID. (e) and (f) Samples of R-CEMEE.

TABLE V
COMPARISON OF THE ANGLE PREDICTION ERRORS OF DIFFERENT LOSS TRAINING METHODS (RADIANS)

Scene	Method	R-SSDD	R-HRSID	CEMEE
Inshore	+ Rotated IOU loss	0.1222	0.1098	0.0651
	+ GPD loss	0.0870	0.1023	0.0542
Offshore	+ Rotated IOU loss	0.0893	0.0548	0.0522
	+ GPD loss	0.0846	0.0530	0.0438

angular error is typically computed using the minimum angular difference

$$\text{Error}_{\text{angle}} = \min\left(\left|\theta_{\text{pred},i} - \theta_{\text{gt},i}\right|, 180^\circ - \left|\theta_{\text{pred},i} - \theta_{\text{gt},i}\right|\right) * (\pi/180). \quad (43)$$

The angle prediction errors of two prediction methods across different scenes on the three datasets are shown in Table V. In the baseline model, an angle prediction vector is added, and the rotated IoU loss [33] is directly calculated for regression, serving as our comparison method. As indicated in Table V, for ships in inshore scenes, the method based on rotated IoU loss yields MAE values of 0.1222, 0.1098, and 0.0651 on the three datasets, respectively. In contrast, our proposed method, which utilizes a Gaussian-based metric combined with ProbIOU loss, results in MAE values of only 0.0870, 0.1023, and 0.0542 across these datasets. The MAE is reduced by 0.0352, 0.0075, and 0.0109 (with angles converted to radians). Regarding offshore ships, the MAE values achieved by our model on the three datasets are 0.0846, 0.0530, and 0.0438, which are significantly lower than 0.0893, 0.0548, and 0.0522 of the comparison method. These quantified angular prediction errors clearly illustrate the effectiveness of our

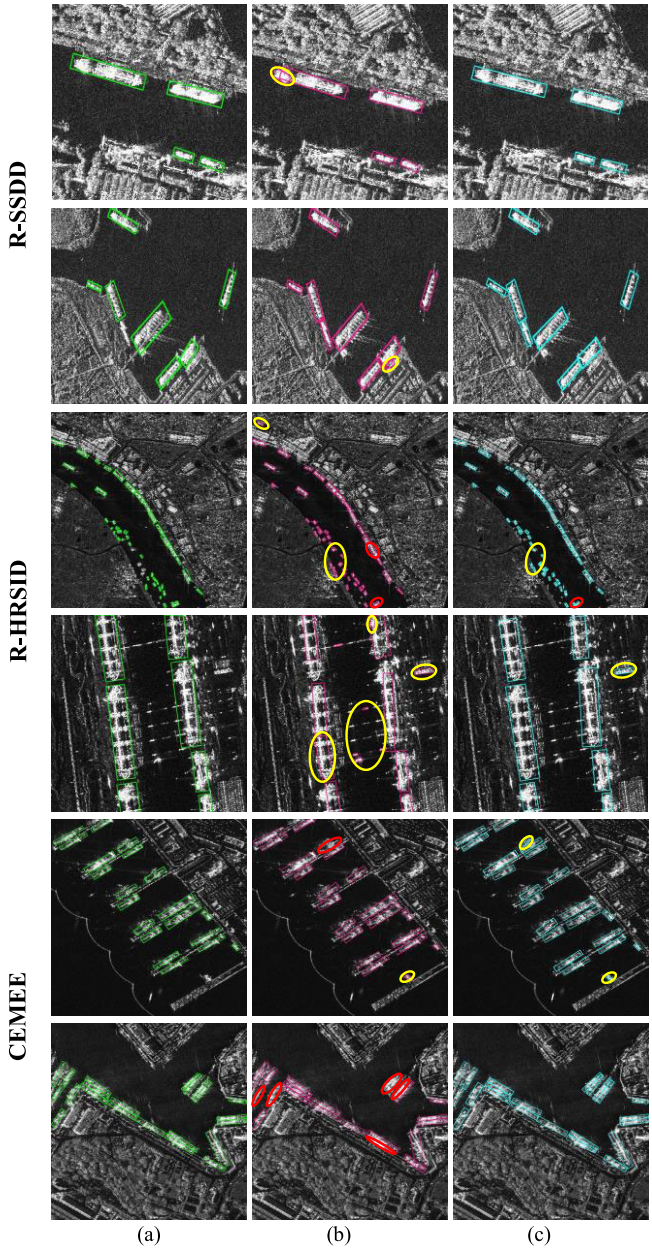


Fig. 11. Assessment of visual detection results of various loss training models. (a) Ground truth. (b) Prediction ships with rotated IoU loss. (c) Prediction ships with GPD loss. A yellow circle indicates false instance and a red circle indicates missed instance.

Gaussian-based angle prediction and loss calculation method, which reduces the computational complexity while enhancing prediction accuracy. In addition, we provide visual detection results for both methods in inshore scenarios across the three datasets, as shown in Fig. 11. In the figure, cyan represents

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TABLE VI
QUANTITATIVE COMPARISON OF DIFFERENT ARBITRARY-DIRECTION SAR SHIP DETECTION ALGORITHMS ON R-SSDD (%)

Type	Method	Entire Scenes				Inshore Scenes				Offshore Scenes			
		P	R	mAP	F1-Score	P	R	mAP	F1-Score	P	R	mAP	F1-Score
Anchor-free	CFA [54]	84.75	88.61	80.89	86.64	65.27	71.85	68.39	68.40	92.53	94.84	90.01	93.67
	Oriented RepPoints [55]	89.46	91.80	89.83	90.61	77.73	80.67	78.07	79.17	93.88	95.94	90.50	94.90
	Rotated RepPoints [56]	86.33	79.84	70.39	82.96	69.83	52.52	47.60	59.95	91.00	90.00	84.88	90.50
	SASM RepPoints [57]	76.67	88.72	80.93	82.26	63.29	76.05	69.45	69.09	81.92	93.44	89.83	87.30
Two-Stage	Faster-RCNN-OBB [18]	90.14	89.52	80.49	89.83	76.89	81.09	77.87	78.93	95.49	92.66	89.91	94.05
	Gliding Vertex [58]	84.80	82.57	78.09	83.67	69.43	66.81	57.36	68.09	90.42	88.44	80.84	89.42
	Oriented R-CNN [59]	92.13	90.66	89.89	91.39	82.79	84.87	79.69	83.82	95.81	92.81	90.39	94.29
	ReDet [60]	90.78	91.91	90.07	91.34	80.25	81.93	78.97	81.08	94.74	95.63	90.49	95.18
	RoI Transformer [61]	92.60	91.23	89.87	91.91	83.05	82.35	79.23	82.70	96.18	94.53	90.46	95.35
One-Stage	ATSS-OBB [62]	91.65	91.23	89.39	91.44	83.41	80.25	78.04	81.80	94.57	95.31	90.20	94.94
	ConvNeXt [63]	90.74	87.02	80.12	88.84	87.18	71.43	69.18	78.52	91.81	92.81	89.32	92.31
	CSL [64]	85.54	79.50	68.78	82.41	72.34	57.14	45.79	63.85	89.49	87.81	79.04	88.64
	FCOS-OBB [65]	93.21	90.66	89.51	91.92	87.10	79.41	70.55	83.08	95.29	94.84	90.38	95.06
	GWD [66]	90.98	84.97	79.58	87.87	84.66	67.23	59.35	74.94	92.87	91.56	89.07	92.21
	KFIoU [67]	89.59	92.14	89.65	90.85	79.51	81.51	78.32	80.50	93.32	96.09	90.52	94.68
	KLD [68]	90.53	90.32	89.11	90.42	79.83	79.83	70.41	79.83	94.51	94.22	90.25	94.36
	PSC [69]	83.27	77.68	67.85	80.38	69.63	55.88	48.00	62.00	87.42	85.87	77.42	86.64
	R3Det [70]	88.84	89.75	80.86	89.29	76.89	76.89	69.16	76.89	93.22	94.53	90.41	93.87
	RetinaNet-OBB [71]	91.65	84.97	79.90	88.18	83.77	67.23	59.23	74.59	94.06	91.56	89.75	92.79
	RTMDet-OBB [49]	91.97	92.60	89.98	92.28	84.32	83.61	80.22	83.96	94.75	95.94	90.22	95.34
	S2ANet [72]	90.43	88.27	87.88	89.34	76.72	74.79	68.10	75.74	95.52	93.28	90.59	94.39
	MSDFF-Net(Ours)	93.95	94.72	**91.55**	**94.33**	86.09	86.38	**82.26**	**86.23**	96.80	97.95	**91.49**	**97.37**

TABLE VII
QUANTITATIVE COMPARISON OF DIFFERENT ARBITRARY-DIRECTION SAR SHIP DETECTION ALGORITHMS ON R-HRSID (%)

Type	Method	Entire Scenes				Inshore Scenes				Offshore Scenes			
		P	R	mAP	F1-Score	P	R	mAP	F1-Score	P	R	mAP	F1-Score
Anchor-free	CFA [54]	77.87	86.12	80.29	81.79	62.10	75.11	67.50	67.99	95.73	96.52	90.68	96.12
	Oriented RepPoints [55]	75.63	86.65	80.24	80.77	58.59	75.50	66.27	65.98	96.13	97.17	90.78	96.65
	Rotated RepPoints [56]	76.05	70.49	66.78	73.16	62.68	57.54	46.67	60.00	88.44	82.71	79.11	85.48
	SASM RepPoints [57]	69.86	81.83	77.91	75.37	60.50	67.00	57.68	63.58	77.80	95.83	90.24	85.88
Two-Stage	Faster-RCNN-OBB [18]	75.43	81.32	78.37	78.26	56.66	65.99	56.06	60.97	96.14	95.79	90.25	95.96
	Gliding Vertex [58]	79.28	79.89	71.54	79.58	62.94	64.25	57.57	63.59	95.11	94.64	90.51	94.87
	Oriented R-CNN [59]	81.51	82.34	80.18	81.92	65.73	65.76	59.79	65.74	96.92	96.29	90.79	96.60
	ReDet [60]	80.36	83.98	80.24	82.13	64.59	70.41	65.56	67.37	96.56	96.78	90.77	96.67
	RoI Transformer [61]	82.46	84.01	80.30	83.23	67.47	70.52	66.28	68.96	97.35	96.75	90.74	97.05
One-Stage	ATSS-OBB [62]	79.29	81.61	78.94	80.43	62.37	66.52	56.57	64.38	96.43	95.86	90.39	96.14
	ConvNeXt [63]	76.73	71.96	68.48	74.27	56.06	48.97	37.23	52.28	93.81	93.66	89.61	93.73
	CSL [64]	79.81	77.89	69.92	78.84	62.13	60.18	52.62	61.14	96.26	94.61	90.07	95.43
	FCOS-OBB [65]	78.59	80.12	78.23	79.35	60.13	63.04	54.13	61.55	97.02	96.25	90.53	96.63
	GWD [66]	83.27	80.47	78.33	81.85	68.21	64.67	55.64	66.39	96.96	95.43	90.38	96.19
	KFIoU [67]	86.80	78.88	71.60	82.65	73.37	61.17	57.10	66.72	97.59	95.60	90.61	96.58
	KLD [68]	80.64	78.67	71.92	79.64	63.14	60.95	56.95	62.03	96.83	95.40	90.80	96.11
	PSC [69]	77.33	77.22	69.43	77.27	57.99	58.94	45.49	58.46	96.22	94.48	89.86	95.34
	R3Det [70]	81.25	80.41	79.39	80.83	66.54	65.26	58.72	65.89	94.90	94.71	90.73	94.80
	RetinaNet-OBB [71]	78.35	78.38	69.74	78.36	59.99	61.05	52.16	60.52	96.29	96.74	89.95	96.51
	RTMDet-OBB [49]	81.69	85.06	80.65	83.34	66.86	72.40	67.58	69.52	96.82	97.01	90.78	96.91
	S2ANet [72]	84.33	85.68	80.91	85.00	70.84	73.58	68.45	72.18	97.62	97.11	90.83	97.36
	MSDFF-Net(Ours)	83.58	88.07	**83.14**	**85.77**	69.67	75.54	**70.12**	**72.49**	98.37	98.02	**91.92**	**98.19**

our detection results, pink illustrates the detection ships with the comparison algorithm, yellow ellipses indicate FPs, and red ellipses highlight the missed detections. From Fig. 11, it is clear that our proposed head prediction method not only provides more accurate OBBs that fit the targets better but also exhibits a significant advantage in overall detection precision–recall rate.

E. Comparison Experiment

To comprehensively showcase the viability of our proposed MSDFF-Net, a thorough comparison with 21 DL-based OBB detection algorithms was conducted in this section. These 21 detection algorithms are categorized into three main groups, including four anchor-free detection methods: CFA [54], Oriented RepPoints [55], Rotated RepPoints [56], and SASM RepPoints [57]; five two-stage detection methods: Faster-RCNN-OBB [18], Gliding Vertex [58], Oriented R-CNN [59], ReDet [60], and RoI Transformer [61]; and 12 single-stage detection methods: ATSS-OBB [62], ConvNeXt [63], CSL [64], FCOS-OBB [65], GWD [66], KFIoU [67], KLD [68], PSC [69], R3Det [70], RetinaNet-OBB [71], RTMDet-OBB [49], and S2ANet [72]. All these algorithms were implemented within the same Pytorch-based mmrotate

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TABLE VIII
QUANTITATIVE COMPARISON OF DIFFERENT ARBITRARY-DIRECTION SAR SHIP DETECTION ALGORITHMS ON CEMEE (%)

Type	Method	Entire Scenes				Inshore Scenes				Offshore Scenes			
		P	R	mAP	F1-Score	P	R	mAP	F1-Score	P	R	mAP	F1-Score
Anchor-free	CFA [54]	67.64	82.15	75.45	74.19	66.50	81.33	74.29	73.17	83.17	92.31	90.29	87.50
	Oriented RepPoints [55]	72.52	85.20	76.60	78.35	71.50	84.76	75.97	77.57	86.84	90.66	89.75	88.71
	Rotated RepPoints [56]	71.76	56.62	47.22	63.30	70.05	54.98	46.45	61.61	91.50	76.92	70.51	83.58
	SASM RepPoints [57]	63.82	69.41	59.15	66.50	63.04	67.78	58.04	65.32	72.12	89.56	81.70	79.90
Two-Stage	Faster-RCNN-OBB [18]	61.75	78.87	63.02	69.27	60.24	77.78	61.74	67.90	83.58	92.31	89.86	87.73
	Gliding Vertex [58]	70.76	77.80	66.23	74.11	69.56	76.89	65.45	73.04	86.63	89.01	80.40	87.80
	Oriented R-CNN [59]	75.02	88.53	79.20	81.22	74.05	88.13	78.96	80.48	88.54	93.41	89.83	90.91
	ReDet [60]	77.82	83.51	77.98	80.56	76.77	82.71	77.31	79.63	91.40	93.41	90.54	92.39
	RoI Transformer [61]	79.20	82.81	78.78	80.96	78.15	82.04	78.22	80.05	92.82	92.31	90.35	92.56
One-Stage	ATSS-OBB [62]	68.94	75.58	66.21	72.11	67.66	74.40	65.17	70.87	85.42	90.11	89.49	87.70
	ConvNeXt [63]	82.65	72.66	67.66	77.33	81.84	71.29	66.86	76.20	91.57	89.56	80.99	90.55
	CSL [64]	72.27	45.76	39.78	56.04	70.64	43.20	38.68	53.61	85.98	77.47	68.67	81.50
	FCOS-OBB [65]	77.93	78.25	69.17	78.09	76.88	77.29	68.73	77.08	91.11	90.11	89.31	90.61
	GWD [66]	82.22	52.30	49.53	63.93	80.99	50.00	48.61	61.83	93.04	80.77	80.22	86.47
	KFIoU [67]	77.35	79.48	69.80	78.40	76.2	78.53	69.23	77.35	92.22	91.21	89.89	91.71
	KLD [68]	78.83	84.54	79.07	81.59	77.87	84.00	78.57	80.82	91.71	91.21	90.35	91.46
	PSC [69]	79.27	46.97	41.65	58.99	77.81	47.38	40.66	58.90	92.21	78.02	72.11	84.52
	R3Det [70]	77.93	77.84	69.80	77.88	76.83	76.80	69.37	76.81	91.67	90.66	89.73	91.16
	RetinaNet-OBB [71]	80.84	51.69	49.07	63.06	79.40	49.16	41.06	60.72	93.21	82.97	80.38	87.79
	RTMDet-OBB [49]	84.38	84.38	79.20	84.38	75.49	83.64	78.70	79.36	88.54	93.41	90.42	90.91
	S2ANet [72]	79.66	77.63	70.32	78.63	78.63	76.71	69.86	77.66	92.57	89.01	81.51	90.76
	MSDFF-Net(Ours)	86.29	87.60	**81.74**	**86.94**	78.24	87.07	**81.34**	**82.42**	90.30	95.45	**92.07**	**92.80**

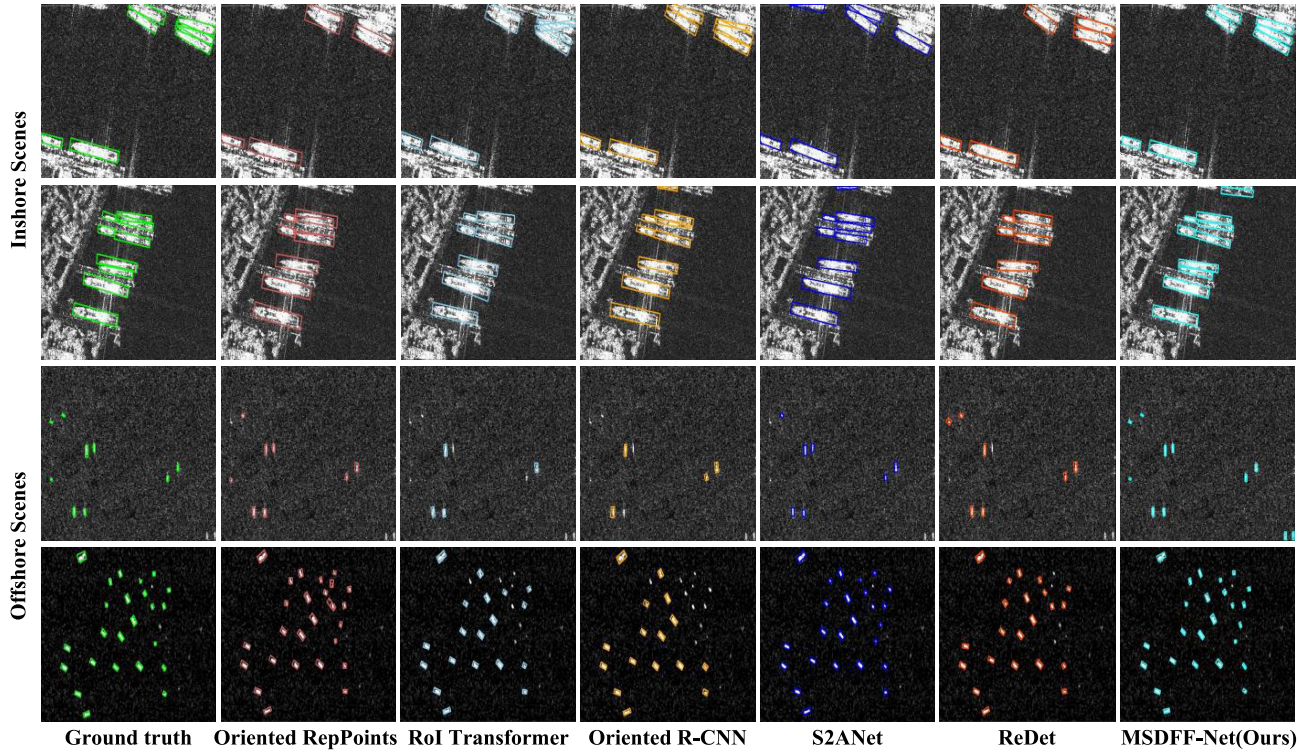


Fig. 12. Visual ship prediction outcomes of various models on the R-SSDD dataset. The top two columns represent the visual results in inshore scenes, while the bottom two columns represent the visual results in offshore scenes. Different colored boxes represent the detection results of different models. (From left to right) Correspond to the ground truth, the results of oriented RepPoints, RoI transformer, oriented R-CNN, S2ANet, ReDet, and the proposed MSDFF-Net, respectively.

framework [52] to ensure the accuracy of the comparisons. In addition, all methods were evaluated under the same software and hardware environment. The comparative analysis will be performed from three perspectives: quantitative results, qualitative results, and P–R curve analysis.

1) Quantitative Results: The P, R, mAP, and $F1$-score of all algorithms across various scenes on the R-SSDD, R-HRSID, and CEMEE datasets are shown in Tables VI–VIII, respectively. (The top results are emphasized in bold, while the next best results are indicated with underlining.)

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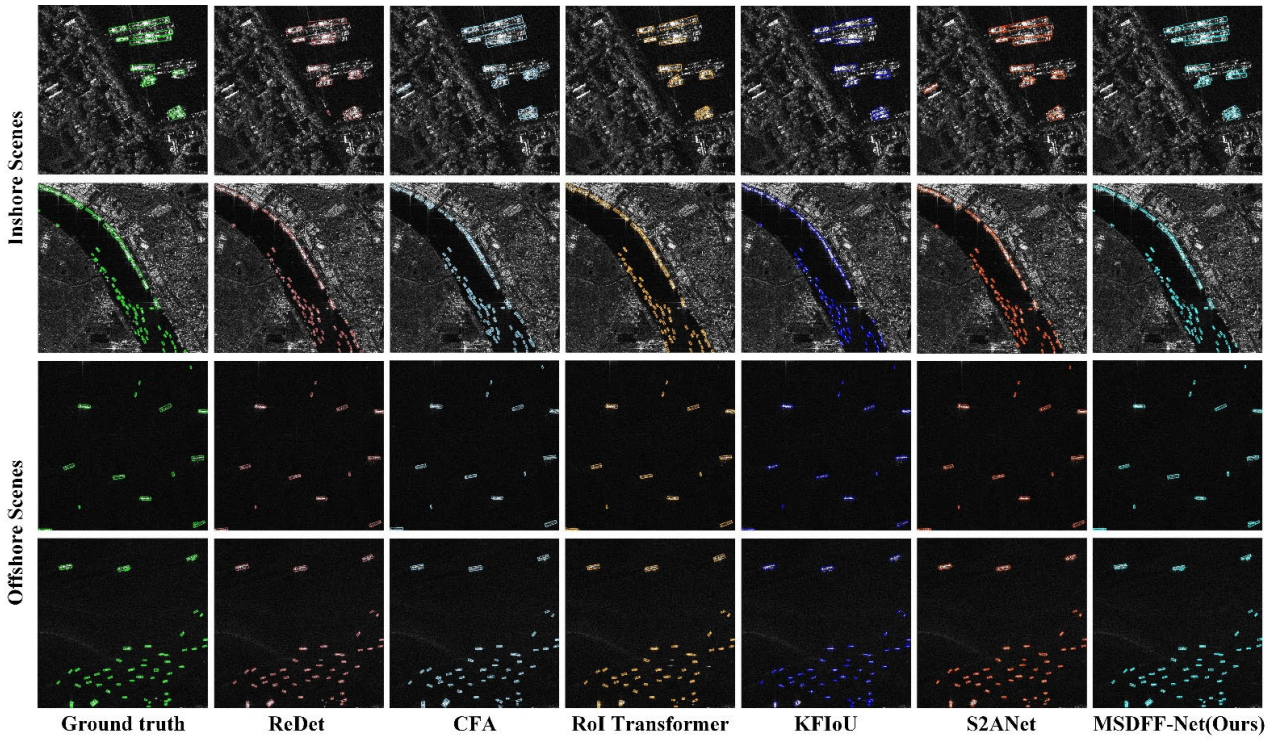


Fig. 13. Visual ship prediction outcomes of various models on the R-HRSID dataset. The top two columns represent the visual results in inshore scenes, while the bottom two columns represent the visual results in offshore scenes. Different colored boxes represent the detection results of different models. (From left to right) Correspond to the ground truth, the results of ReDet, CFA, RoI transformer, KFIoU, S2ANet, and the proposed MSDFF-Net, respectively.

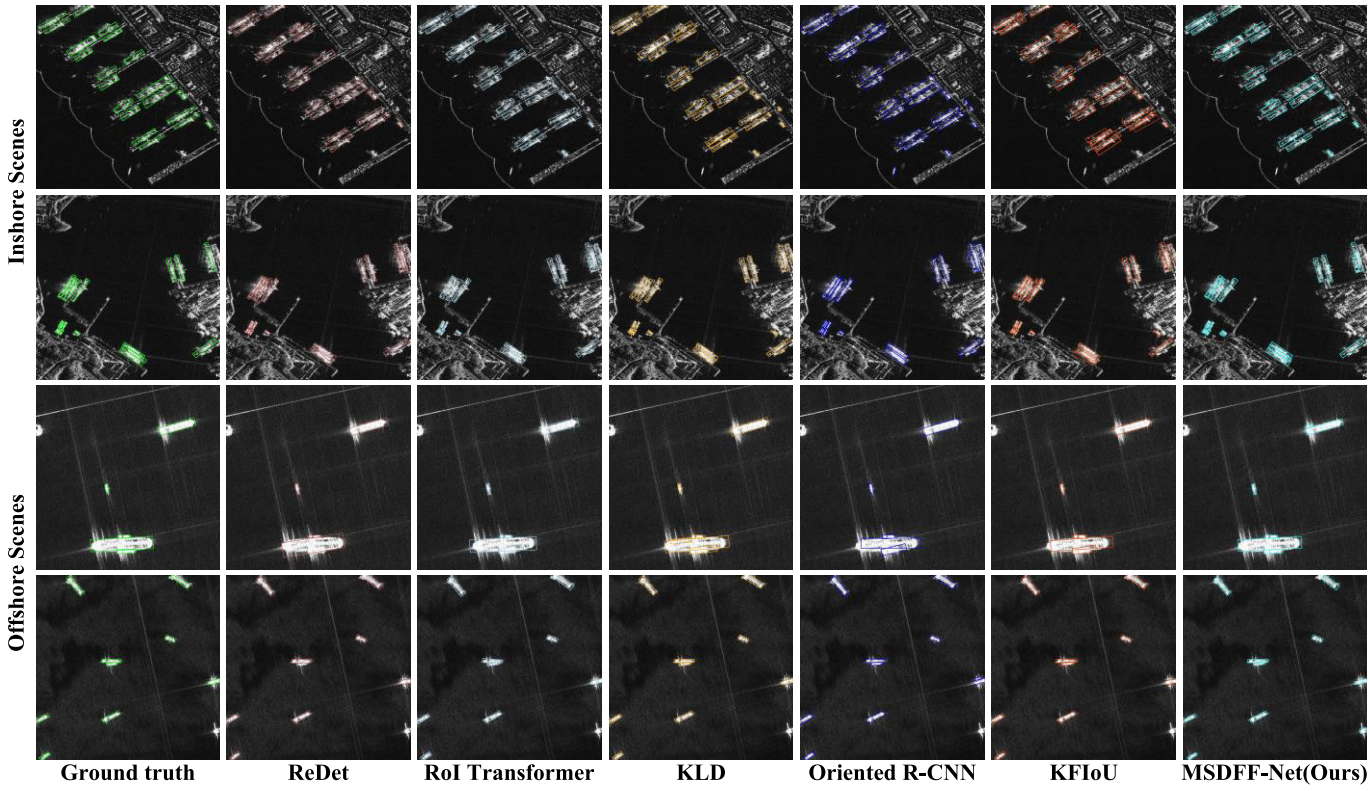


Fig. 14. Visual ship prediction outcomes of various models on the CEMEE dataset. The top two columns represent the visual results in inshore scenes, while the bottom two columns represent the visual results in offshore scenes. Different colored boxes represent the detection results of different models. (From left to right) Correspond to the ground truth, the results of ReDet, RoI Transformer, KLD, oriented R-CNN, KFIoU, and the proposed MSDFF-Net, respectively.

From Table VI, it is evident that MSDFF-Net (93.95% precision, 94.72% recall, 91.55% mAP, and 94.33% $F1$-score) outperforms other methods in the overall scenes of the widely used R-SSDD dataset. The suboptimal method achieves

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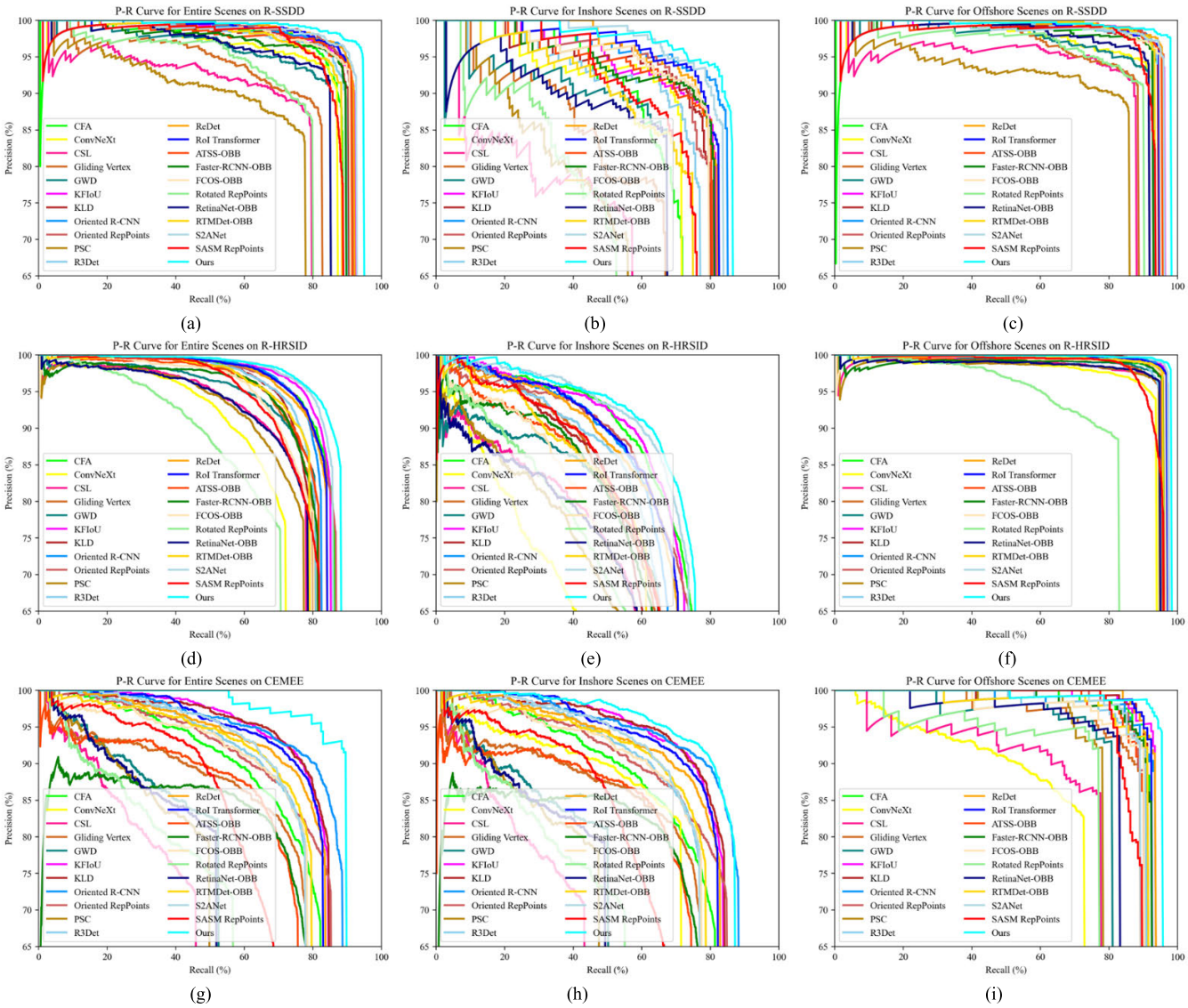


Fig. 15. P–R curves for various models on the experiment datasets. (a)–(c) P–R curves for R-SSDD detection results. (d)–(f) Detection results for R-HRSID. (g)–(i) Detection results for CEMEE. (From left to right) Curves correspond to the entire scene, inshore scene, and offshore scene, respectively.

90.07% mAP and 92.28% $F1$-score, which are 1.48% and 2.05% lower than those of our MSDFF-Net. In the inshore scenes, MSDFF-Net surpasses all other methods, with a precision and recall of 86.09% and 86.38%, and an overall mAP improvement of approximately 2.04% over the suboptimal method. This significant performance boost shows that MSDFF-Net effectively improves the model's ability to handle multiscale ships in complex and dynamic environments, leading to stronger multiscale feature representation under challenging conditions. For offshore scenes, our method continues to surpass other approaches, demonstrating that, while focusing on improving ship detection in more intricate scenarios, we have not neglected the feature extraction of offshore ship targets. Table VII shows that although S2ANet (84.33% precision, 85.68% recall, 80.91% mAP, and 85.00% $F1$-score) performs strongly on this dataset, MSDFF-Net (83.58% precision, 88.07% recall, 83.14% mAP, and 85.77% $F1$-score) still holds a significant advantage in overall detection performance. The results in Table VIII demonstrate that MSDFF-Net (86.29% precision, 87.60% recall, 81.74% mAP, and 86.94% $F1$-score) overwhelmingly surpasses the other 21 algorithms. From the combined results of all three tables, it is clear that the proposed MSDFF-Net exhibits strong robustness across different datasets, fully validating the rationality of the improved modules and significantly enhancing SAR ship detection performance across images with varying resolutions.

2) Qualitative Results: Figs. 12–14 show the qualitative detection results of the top-10 algorithms on the SSDD, HRSID, and CEMEE datasets, respectively. The first three columns of each figure are the detection results of nearshore scenes, and the last three columns are the detection outcomes for offshore scenes. Various methods are depicted in distinct colors (to obtain the original high-definition images, kindly reach out to the author via email). By examining the visual outcomes, it can be found that the proposed MSDFF-Net

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demonstrates excellent performance in complex scenes and exhibits strong multiscale detection capabilities. At the same time, for dense small-scale ships, our proposed method can almost completely achieve detection, except for a very small number of targets that are missed. This shows that MSDFF-Net can effectively handle a variety of changing scenes. However, in certain particularly challenging scenes, some missed detections still occur, likely due to severe scene interference that significantly hinders target feature extraction, one of the most challenging aspects of SAR ship detection.

3) P–R Curve Analysis: The P–R curves of 22 methods on three different datasets are presented in Fig. 15. The P–R curve can intuitively show the comprehensive detection performance of different methods. The region beneath P–R curve represents an mAP index of the prediction method, and the intersection of the curve with the x- and y-axes, respectively correspond to the detection rate along with the recall rate. From Fig. 15, we can more intuitively see the superior comprehensive detection performance of our proposed MSDFF-Net (cyan blue curve). Based on Fig. 15(a)–(c), our proposed MSDFF-Net demonstrates robust detection performance across various scenes in the classic SSDD dataset. Notably, in nearshore scenes, the detection recall rate is significantly higher than that of other models. In addition, as shown in Fig. 15(d)–(f), our proposed algorithm exhibits strong generalization capabilities across different datasets. For the CEMEE dataset, which comprises a high proportion of complex scenes, Fig. 15(g)–(i) indicates that our algorithm achieves optimal overall detection performance. This improvement is attributed to the targeted solutions we designed to address existing imbalance problems.

V. Discussion

In this section, we first compare and analyze the model complexity and computational efficiency, then analyze the improvement effect of the model's attention area in combination with feature map visualization, and finally discuss the application effect of the proposed algorithm on large-scene images.

A. Complexity and Efficiency Analysis

Model complexity and computing efficiency are one of the important indicators to measure the detection model. To this end, we calculated the parameter (Para), FLOP amount, and model weights (Weights) and calculated FPS value of all 22 methods on the R-SSDD dataset, which is presented in Table IX. In terms of model complexity, our MSDFF-Net (8.94 M Para, 9.57 G FLOPs, 143.46-Mb weights, and 135.79 FPS) is second only to RTMDet-OBB (8.86 M Para, 9.40 G FLOPs, 141.26-Mb weights, and 139.28 FPS), but is significantly superior to the other 20 models. Compared with RTMDet-OBB, MSDFF-Net only increases the number of parameters by 0.08 M, FLOPs by 0.17, and weights by 2.20 Mb. In terms of computational efficiency, although the anchor-free algorithm does not require preset anchor boxes, the heavy target point set regression does not bring a speed advantage. Meanwhile, one-stage approaches generally surpass two-stage methods in overall performance. Our MSDFF-Net achieves 135.79 FPS, second only to RTMDet-OBB at 139.28 FPS, and significantly outperforms the other 20 models. In conclusion, the proposed method significantly enhances the overall detection performance of the model while maintaining a minimal increase in the model complexity and computing time.

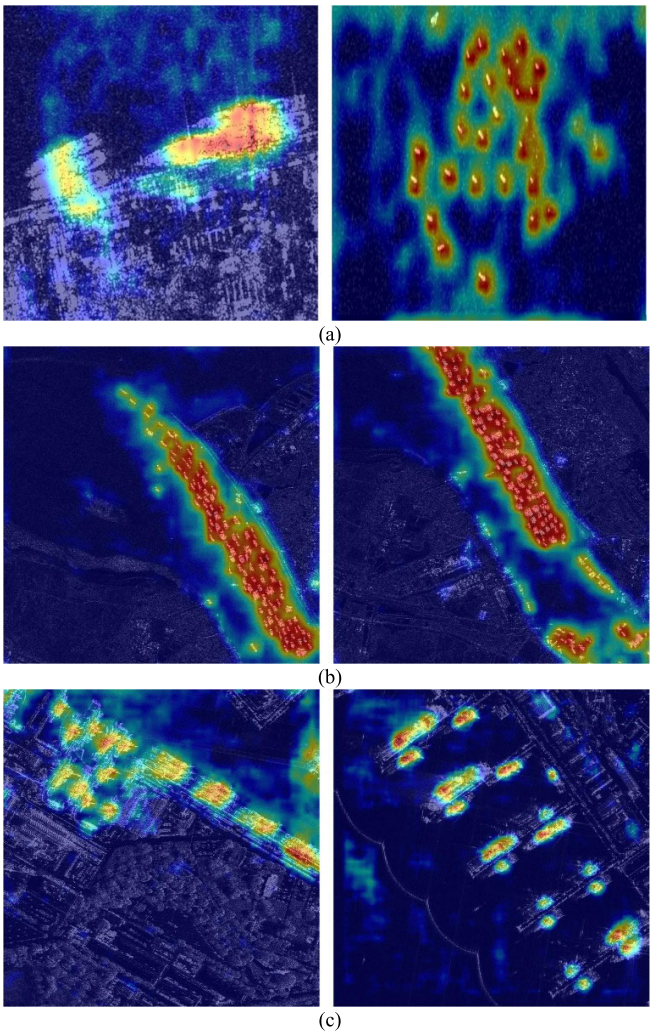


Fig. 16. Visualization results of Eigen-CAM features in final output layer of our designed model. (a) Visualization features of the R-SSDD dataset. (b) Visualization features of the R-HRSID dataset. (c) Visualization features of the CEMEE dataset.

B. Eigen-CAM Feature Visualization

To further analyze and validate the regions of interest within the proposed network's overall feature extraction, we used Eigen-CAM to visualize the output features, which is presented in Fig. 16. By presenting the visualization results of our proposed method across three datasets, we assess the network's capability to capture crucial regions that influence detection performance. From the Eigen-CAM visualizations, it is evident that for nearshore scenarios, our MSDFF-Net model effectively captures the primary superstructures of closely docked ships and distinguishes between unique feature regions of different vessels, enabling the model to specifically

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TABLE IX
COMPARISON OF MODEL COMPLEXITY AND COMPUTATIONAL EFFICIENCY WITH DIFFERENT ARBITRARY-DIRECTION SAR SHIP DETECTION ALGORITHMS ON R-SSDD

Type	Method	Para(M)	FLOPs(G)	Weights(Mb)	FPS
Anchor-free	CFA [38]	36.60	48.56	280.51	29.46
	Oriented RepPoints [39]	36.60	48.56	280.51	28.55
	Rotated RepPoints [40]	36.60	48.56	280.51	31.45
	SASM RepPoints [41]	36.60	48.56	280.51	31.38
Two-Stage	Faster-RCNN-OBB [13]	41.12	50.36	314.99	34.51
	Gliding Vertex [42]	41.13	50.37	315.02	33.83
	Oriented R-CNN [43]	41.13	50.40	315.00	30.19
	ReDet [44]	31.54	15.11	245.01	18.47
	RoI Transformer [45]	55.03	51.38	421.06	26.43
One-Stage	ATSS-OBB [46]	36.01	51.79	276.01	37.17
	ConvNeXt [47]	37.14	53.38	425.29	37.75
	CSL [48]	37.06	57.49	283.97	33.81
	FCOS-OBB [49]	31.89	51.55	244.55	38.69
	GWD [50]	36.13	52.39	276.85	36.92
	KFIoU [51]	41.58	82.17	318.50	26.84
	KLD [52]	39.22	69.29	300.48	29.58
	PSC [53]	36.25	53.07	278.71	30.78
	R3Det [54]	41.58	82.17	282.47	18.77
	RetinaNet-OBB [55]	36.13	52.39	276.85	36.22
	RTMDet-OBB [36]	**8.86**	**9.40**	**141.26**	**139.28**
	S2ANet [56]	38.54	49.05	295.25	29.86
	MSDFF-Net(Ours)	8.94	9.57	143.46	135.79

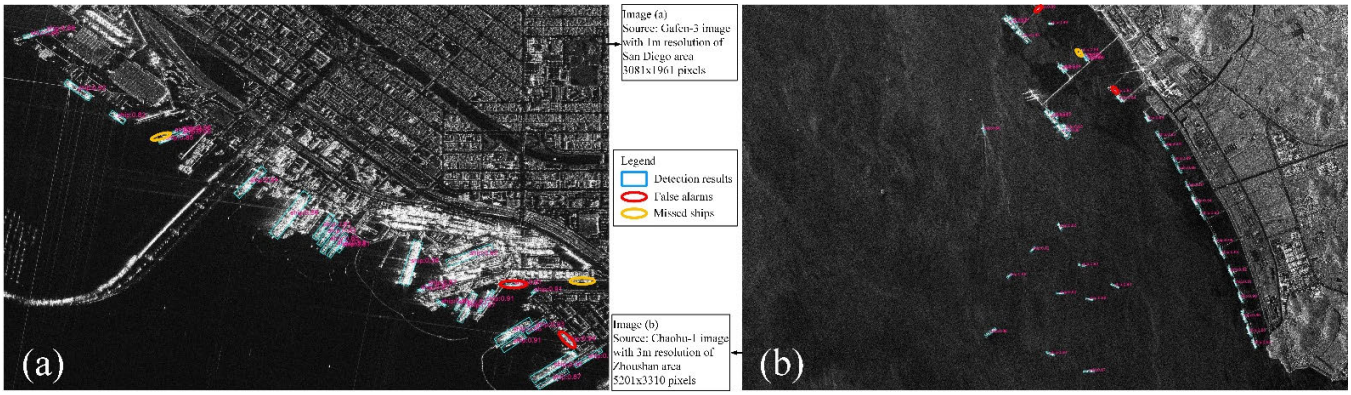


Fig. 17. Visual detection results of our MSDFF-Net on two large-scene images. (a) Image 1. (b) Image2. Cyan indicates our detection results, red represents the false alarms, and yellow denotes the missed ships.

learn location-oriented features of the target areas. Moreover, the regions of interest in the Eigen-CAM visualization of our MGSFA-Net are more concentrated and better aligned with the target contours. For offshore scenes, the model effectively captures key regions of densely distributed small-scale ships without terrestrial noise, showcasing the robust feature extraction and integration capabilities of our method.

C. Validation on Large-Scene SAR Images

We further employed MSDFF-Net to detect ships in two additional large-scale SAR images to validate its robust transferability. The model was trained on 1024 × 1024 image patches cropped from large-scale scenes. During testing, each large-scale image was divided into patches, which were fed into the detection model to obtain localized results. These individual patch results were then merged to produce the final detection outcome for the large scene. Fig. 17 illustrates the corresponding visual detection results. Data from a 1-m resolution Gaofen-3 image of a region in San Diego, with dimensions of 3081 × 1961 pixels, is displayed in Fig. 17(a). Fig. 17(b) illustrates a 3-m resolution image from Chaohu-1, covering a region in Zhoushan, with a dimension of 5201 × 3310 pixels. In the results, cyan denotes our detection results, red indicates FPs, and yellow highlights missed ships. As shown in Fig. 17(a), our method accurately detected 30 ships, with only two FPs and two missed detections. For the second large-scale image, we correctly identified 44 of 45 ships, with only two FPs and one missed detection. These results demonstrate MSDFF-Net's strong transferability and its potential for application in wide-area maritime ship detection in real-world scenarios.

VI. CONCLUSION

In this article, a precise and efficient network, termed MSDFF-Net, was proposed for arbitrary-oriented SAR ship detection. The model is designed to address multiscale imbalance problems, including imbalance in multiscale feature extraction, imbalance in multiscale feature utilization, and imbalance in multiscale target loss regression. To enhance

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feature representation and achieve balanced extraction of multiscale and multilevel ship features, the MSLK-Block was constructed, leveraging adaptive large-kernel convolutions. By implementing the DFF-Block, our model strengthens its capacity to perceive complex environments, thereby balancing spatial and channel-level feature utilization. In addition, an iterative training approach based on GPD measures was applied to mitigate regression loss imbalance in prediction tasks, leading to improved prediction accuracy. Experimental results on the R-SSDD, R-HRSID, and CEMEE datasets confirmed MSDFF-Net's ability to achieve an optimal balance between speed and accuracy in arbitrary-oriented ship detection, making it suitable for real-world applications. Future work will focus on developing integrated detection and recognition frameworks for arbitrary orientations to address sample imbalance challenges in the SAR ship recognition.

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